BONNEVILLE POWER ADMINISTRATION

Effects of Summer Flow Augmentation on the Migratory Behavior and Survival of Juvenile Snake River Fall Chinook Salmon

Annual Report 2004 - 2005 March 2006 DOE/BP-00005362-4



This Document should be cited as follows:

Tiffan, Kenneth, William Connor, "Effects of Summer Flow Augmentation on the Migratory Behavior and Survival of Juvenile Snake River Fall Chinook Salmon", 2004-2005 Annual Report, Project No. 199102900 (et al.), 80 electronic pages, (BPA Report DOE/BP-00005362-4)

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P.O. Box 3621
Portland, OR 97208

This report was funded by the Bonneville Power Administration (BPA), U.S. Department of Energy, as part of BPA's program to protect, mitigate, and enhance fish and wildlife affected by the development and operation of hydroelectric facilities on the Columbia River and its tributaries. The views in this report are the author's and do not necessarily represent the views of BPA.

Annual Report 2004

EFFECTS OF SUMMER FLOW AUGMENATION ON THE MIGRATORY BEHAVIOR AND SURVIVAL OF JUVENILE SNAKE RIVER FALL CHINOOK SALMON

March 2006



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BONNEVILLE POWER ADMINISTRATION

EFFECTS OF SUMMER FLOW AUGMENATION ON THE MIGRATORY BEHAVIOR AND SURVIVAL OF JUVENILE SNAKE RIVER FALL CHINOOK SALMON

ANNUAL REPORT 2004

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Project Number 1991-02900
Contract Number DE-AI79-91BP21708

http://www.efw.bpa.gov/searchpublications/

March 2006

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Executive Summary

This report summarizes results of research activities conducted in 2004 and years previous to aid in the management and recovery of fall Chinook salmon *Oncorhynchus tshawytscha* in the Columbia River basin. For detailed summaries, we refer the reader to the abstracts given on the second page of each chapter. The Annual Reporting section includes information provided to fishery managers in-season and post-season, and it contains a detailed summary of life history and survival statistics on wild Snake River fall Chinook salmon juveniles for the years 1992-2004. Publication is a high priority of our staff. Publication provides our results to a wide audience, and it insures that our work meets high scientific standards. The Bibliography of Published Journal Articles section provides citations for peer-reviewed papers co-authored by personnel of project 1991-02900 that were written or published from 1998 to 2005.

Acknowledgments

We thank our colleagues at the Bonneville Power Administration, Environmental Protection Agency, Fish Passage Center, Idaho Fish and Game, Idaho Power Company, National Marine Fisheries Service, Nez Perce Tribe, Oregon Department of Fish and Wildlife, Pacific States Marine Fisheries Commission, U.S. Army Corps of Engineers, U.S. Fish and Wildlife Service, U.S. Geological Survey, University of Idaho, and Washington Department of Fish and Wildlife. Special thanks to John Yearsley, David Benner, Ben Cope, and Ronnie Mock.

Annual Reporting

CHAPTER ONE

Data and Analyses on Wild Fall Chinook Salmon Subyearlings
1992—2004

by

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Abstract.—Emergence and shoreline rearing of wild fall Chinook salmon subyearlings in the Snake River progressed on an early time schedule in 2004. Growth was slightly lower than average, but the subyearlings still passed Lower Granite Dam on an earlier than average time schedule. The forecast of passage at Lower Granite Dam provided in season to managers in 2004 suggested that passage would be earlier than in most years and it was. The model forecasted the median date of passage 3 d earlier than observed, but it also forecasted 95% passage 14 d earlier than observed. The joint probability of active migration and survival to the tailrace of Lower Granite Dam in 2004 was 45%; 5 percentage points lower than the 1998—2003 median of 49%. This joint probability during 1998—2004 was directly proportional to fork length at release and flow as well as inversely proportional to temperature. A model was fitted from these variables and used to predict this joint probability under the scenarios of no flow augmentation, flow augmentation from July to mid September (i.e., observed), and July-August. Predictions made with the fitted model suggested that summer flow augmentation increased the joint probability of active migration and survival of wild fall Chinook salmon subyearlings of Snake River origin by 1—8 percentage points in 2004. Moreover, modeling results indicated that saving water during July and August for release in September to protect Clearwater River smolts decreased the joint probability of active migration and survival of wild Snake River smolts by only 2 percentage points or less.

Introduction

Wild fall Chinook salmon *Oncorhynchus tshawytscha* subyearlings of Snake River origin listed under the Endangered Species Act (NMFS 1992) migrate seaward in the lower Snake River during late spring and summer when flow is low and spend prolonged periods in Lower Granite Reservoir (Connor et al. 2002, 2003a; Figure 1). During recovery planning, it was determined that summer water conditions in Lower Granite Reservoir were unfavorable for survival (NMFS 1995). In July of 1992, a small volume of stored reservoir water was released to increase flow and decrease water temperature in Lower Granite Reservoir. Thereafter, larger volumes of water were released annually between 5 July and 31 August from Dworshak Reservoir, U.S. Bureau of Reclamation (USBR) projects in southern Idaho, and Brownlee Reservoir (Figure 1). Releasing this stored water is called summer flow augmentation.

There is not enough stored reservoir water available to optimize passage conditions in Lower Granite Reservoir throughout the 5 July to 31 August time period. Consequently fish and water managers meet in the forums of the Fish Passage Advisory Committee (FPAC) and the Technical Management Team (TMT) to formulate a management plan. A 2000 decision by the State of Idaho to deny waivers for exceeding dissolved gas levels above 110% in the tailrace of Dworshak Dam reduced the complexity of the water management plan and the ability of fish and water managers to shape summer flows. To avoid exceeding this "gas cap," the maximum volume of water that can be released from Dworshak Dam was reduced to approximately 397 $m^3/s/d$ (14,000 $ft^3/s/d$). Thus in some years, the full 1.2 million acre feet of stored water in Dworshak Reservoir available for summer flow augmentation cannot be released by the end of August. In 2003, fishery managers decided to release less than 397 m^3/s during periods in early July and late August so that flows could be augmented the first two weeks of September. The assumed benefits of September releases were increased survival of juvenile fall Chinook salmon that hadn't passed Lower Granite Dam by the end of August, and increased survival of immigrating adult fall Chinook salmon and steelhead *O. mykiss*. The mechanism for this increased survival was the prevention of rapid warming in Lower Granite Reservoir when summer flow augmentation was discontinued at the end of August.

Since 1992, personnel of project 199102900 have been assisting fish and water managers during planning, implementation, monitoring, and evaluation of summer flow augmentation. Much of the data provided to fishery managers are in the form of records compiled on wild fall Chinook salmon subyearlings that were implanted with Passive Integrated Transponder (PIT) tags (Prentice et al. 1990a). The PIT-tag data are provided weekly each year via the PIT-tag Information System, where in turn, it is downloaded, analyzed, and posted in real time on Internet services (e.g., DART; Program RealTime, Burgess and Skalski 2001) to allow managers to track the progress of the smolt migration.

In 2004, personnel of project 199102900 provided members of the FPAC and TMT an in-season update in July that included: 1) a summary of 2004 catch and tagging data; 2) a comparison of observed time of fry and parr presence among the years 1992 to 2004; 3) a comparison of parr growth among the years 1992 to 2004; and 4) a forecast of passage at Lower Granite Dam (Figure 1) for wild fall Chinook salmon subyearlings of Snake River origin. The

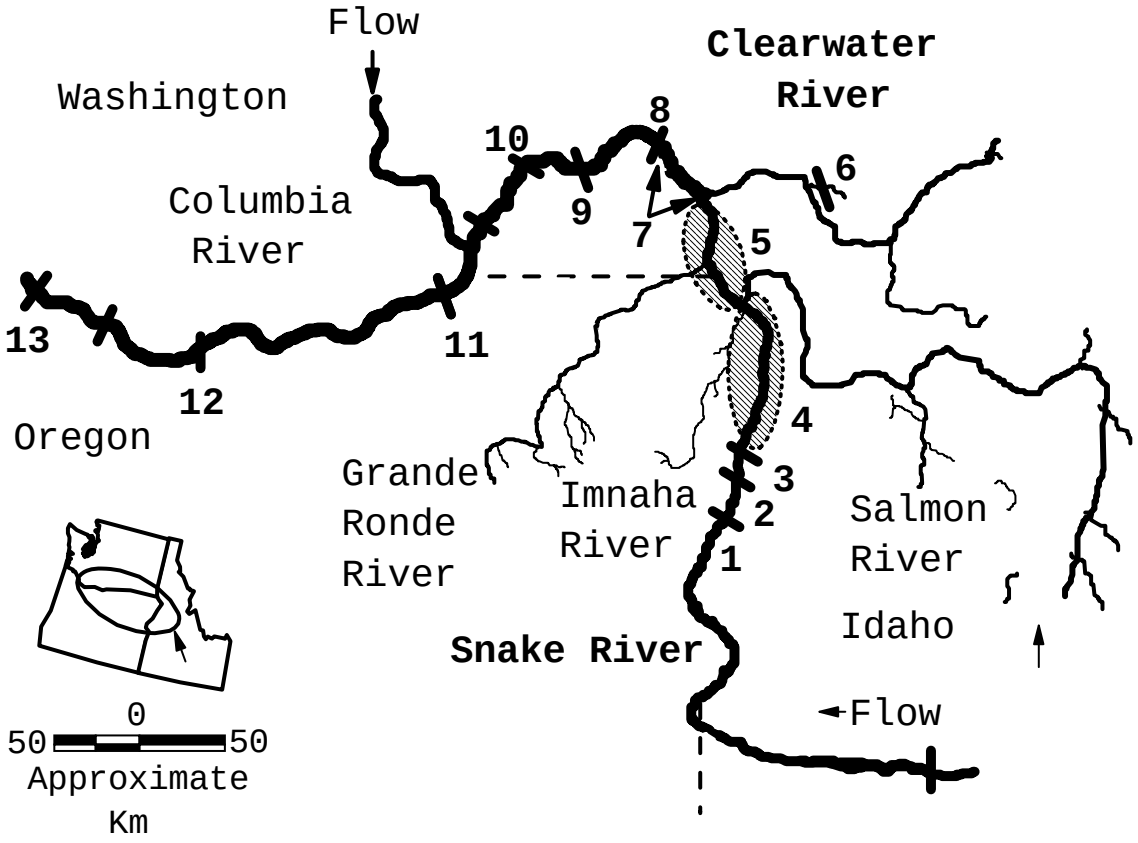


Figure 1.—Locations of the upper and lower reaches of the Snake River where adult fall Chinook salmon spawn and their offspring were captured by using a beach seine (cross-hatched ellipses) and dams where PIT-tagged smolts were detected during seaward migration. The locations are as follows: 1 = Brownlee Reservoir; 2 = Oxbow Reservoir; 3 = Hells Canyon Reservoir; 4 = Snake River upper reach; 5 = Snake River lower reach; 6 = Dworshak Reservoir; 7 = Lower Granite Reservoir; 8 = Lower Granite Dam; 9 = Little Goose Dam; 10 = Lower Monumental Dam; 11 = McNary Dam; 12 = John Day Dam; and 13 = Bonneville Dam.

TMT was also provided with a post-season briefing in late October, 2004 that included a review of the factors that affect migrational behavior and survival of fall Chinook salmon juveniles.

In this chapter, in-season and post-season data collected in 2004 are summarized as follows. Early life history timing and growth in 2004 is described and compared to other years. The passage forecast for PIT-tagged subyearlings at Lower Granite Dam is evaluated. The efficacy of summer flow augmentation during 2004 is assessed with focus on the survival effect of saving some Dworshak Reservoir water for release in September.

Methods

Data collection

The Snake River can be divided into two reaches based on differences in water temperature (Connor et al. 2002). The upper reach extends from the Salmon River confluence to Hells Canyon Dam, and the lower reach extends from the upper end of Lower Granite Reservoir to the Salmon River confluence (Figure 1). Juvenile fall Chinook salmon were captured in these two reaches by use of a beach seine. Sampling began in the upstream reach in 1995, and in the lower reach in 1992. Beach seining typically started in April soon after fry began emerging from the gravel, and was conducted weekly at permanent stations within each reach. From 1992 to 1999, additional non-permanent stations were sampled for three consecutive weeks once a majority of fish was at least 60 mm fork length. Only permanent stations were sampled after 1999. We discontinued all sampling in June or July when beach seine catch fell to near zero indicating that the large majority of fish had moved into Lower Granite Reservoir or to points further downstream.

Field staff implanted PIT tags into parr 60-mm fork length and longer. Tagged parr were released at point of capture after a 15-min recovery period. Some of the PIT-tagged fish were detected as smolts as they passed downstream in the juvenile bypass systems of dams equipped with PIT-tag monitors (Matthews et al. 1977; Prentice et al. 1990b). Operation schedules for the fish bypass systems varied by dam and year. Most of the detections were in the fish bypass systems of Lower Granite, Little Goose, and Lower Monumental dams operated from early April to early November, and at McNary Dam (Figure 1) operated from early April to early December.

Starting in 1999, hatchery fall Chinook salmon subyearlings with in tact adipose fins were released into the Snake River by personnel of the Nez Perce Tribe to supplement wild production. Field personnel subjectively identified these fish in the beach seine catch based on body morphology. Hatchery fish were more slender and appeared to have larger eyes than wild fish. Fish judged to be of hatchery-origin were released back to the river and were not included in analyses.

Accuracy of origin classification

To assess the ability of field personnel to distinguish between hatchery and wild fall Chinook salmon juveniles, every fish was examined for PIT-tag wounds and scars after

removing it from the seine to determine if it was a recaptured PIT-tagged fish. The origin of each recaptured PIT-tagged fish was judged, and then the fish was scanned for its unique tag code. The PIT-tag code was used to trace actual origin. Within-origin classification accuracy (hatchery and wild, separately) and across-origin classification accuracy (hatchery and wild, combined) was calculated. Within-origin classification accuracy was the number of correct classifications divided by the number of recaptured hatchery or wild fish. Across-origin classification accuracy was calculated as the weighted average of the within-run classification accuracy estimates.

Early life history and growth

Capture dates of wild fall Chinook salmon smaller than 45-mm fork length were used to describe time of presence for newly emergent fry (hereafter emergence timing). Capture dates for fish 45-mm fork length and longer to describe time of presence for fall Chinook salmon parr (hereafter, rearing timing). All capture dates were adjusted to Sunday's date the week of sampling to account for differences in day of sampling between the upper and lower reaches of the Snake River. For example, a capture date of 2 May, 1993 (Sunday) was reported for fry and parr collected from 4 May to 6 May (Tuesday to Thursday).

Absolute growth rate (mm/d) during shoreline rearing was calculated using length data from PIT-tagged parr recaptured by beach seine after initial capture and tagging (e.g., Connor and Burge 2003). Absolute growth rate was calculated as: fork length at recapture minus fork length at initial capture divided by the number of days between initial capture and recapture.

The PIT-tag detection data collect at Lower Granite Dam were used to represent the onset of active seaward migration (e.g., Connor et al. 2003a) by subyearling smolts.

Passage forecasting

See Connor et al. (2000) and Connor (2003) for a description of the forecasting methods developed with data collected during 1993—2002. A discriminant function fit from data collected in 1994, 1995, and 1999 was used to predict survival of each PIT-tagged fish. A multiple regression equation fit from the 1994, 1995, and 1999 data was used to predict rate of seaward movement of the survivors measured from release in the Snake River to passage at Lower Granite Dam. A value of 173 (i.e., the location of Lower Granite Dam in kms from the river mouth) was subtracted from the distance from the Snake River mouth where each predicted survivor was initially released. This difference was divided by the predicted rate of seaward movement of the fish. This quotient was added to the initial date of release to estimate a passage date at Lower Granite Dam for each predicted survivor. The estimated passage dates were then accumulated to produce a cumulative frequency distribution.

Forecast performance was assessed three ways. First, the difference between the forecasted and observed dates of 50% and 95% passage were calculated. Second, the absolute difference between observed and forecasted daily cumulative (%) passage during 1 July-31 August was calculated. The average of these daily difference was named the "mean daily error rate". Mean daily error rates calculated for July and August were used to assess forecast

accuracy. Third, the number of days the forecast and its 90% intervals failed to contain observed passage during July and August was used to assess precision.

Active migration and survival

See Connor et al. (2003b) for details on the following analyses. The annual samples of PIT-tagged fall Chinook salmon were divided into four sequential within-year release groups referred to as "cohorts." The single release-recapture model (Cormack 1964; Skalski et al. 1998) was used to estimate the joint probability of active migration and survival ($\pm$ SE) to the tailrace of Lower Granite Dam for each cohort.

Hereafter, I clarify by example what is meant by "the joint probability of active migration and survival." Wild subyearling fall Chinook salmon from cohort 1 emerge and migrate earlier than those fish from cohorts 2—4 (Connor et al. 2003b). Wild fall Chinook salmon from cohort 1 rarely exhibit a reservoir-type life history. A typical "survival" estimate to the tailrace of Lower Granite Dam for cohort 1 made by use of the single release-recapture model might be 69%. In reality, this estimate is a product of the probability of successfully becoming a subyearling smolt and actively migrating (e.g., 98%) and the probability of surviving to the tailrace of Lower Granite Dam (e.g., 70%). That is, 98% x 70% = 69%. In this example, the survival probability estimate for cohort 1 is only 1 percentage point lower than true survival. Therefore, the estimate is a relatively unbiased measure of survival. Wild subyearling fall Chinook salmon from cohort 4 emerge and begin seaward movement later in the year than those from cohort 1 and are more likely to cease active migration and exhibit the reservoir-type life history (Connor et al. 2005). A typical survival estimate to the tailrace of Lower Granite Dam for cohort 4 might be 16%. As in the first example, this estimate is a product of the probability of successfully becoming a subyearling smolt and actively migrating (e.g., 40%) and the probability of surviving to the tailrace of Lower Granite Dam (e.g., 40%). That is, 40% x 40% = 16%. The survival probability estimate in this example is 24 percentage points lower than true survival. Therefore, the survival estimate is inaccurate and best portrays the joint probability of active migration and survival. To account for the potential bias in survival probability estimates I use the phrase "the joint probability of active migration and survival" instead of the probability of survival.

The efficacy of flow augmentation

To accomplish this objective, an ordinary-least squares multiple regression model was fitted to predict the joint probability of active migration and survival to the tailrace of Lower Granite Dam. This joint probability for each of four the 1998—2004 cohorts provided values for the dependent variable. The predictor variables were the median day of year of release, mean fork length (mm) at release, a flow (m^3/s) exposure index calculated as the mean outflow measured at Lower Granite Dam during the period when the majority of smolts passed the dam, and a water temperature (°C) exposure index calculated as the mean temperature measured in the tailrace of Lower Granite Dam during the period when the majority of smolts passed the dam. See Connor et al. (2003b) for more details on calculating exposure indices.

A product moment correlation coefficient (r) was calculated between survival and each independent variable. Variables that were significantly ($P < 0.05$) correlated with the joint probability of active migration and survival were entered into a starting regression model along with all possible interaction terms. The final model had a Mallow's Cp score most similar to the number of model parameters (e.g., B_o, B_1, $B_3 = 3$), the lowest AIC value, and a high R^2 value.

The final model was used to predict the joint probability of active migration and survival (including 95% lower and upper confidence limits) under three scenarios, namely, baseline, observed, and July-August. The baseline scenario approximated the flow volumes and daily mean temperatures in the tailrace of Lower Granite Dam had no stored reservoir water been released from Dworshak Reservoir (about 1.2 million acre feet) or USBR projects (approximately 342,000 acre feet) and Brownlee Reservoir (approximately 202,000 acre feet) combined. The observed scenario included the actual daily mean flow volumes and tailrace temperatures measured at Lower Granite Dam in 2004 that resulted from releasing stored water in Dworshak Reservoir from July to early September. The July-August scenario approximated the daily mean flow volumes and temperatures in tailrace of Lower Granite Dam had the 1.2 million acre feet of stored water in Dworshak Reservoir been released at a rate of 397 $m^3/s/d$ (14,000 $ft^3/s/d$) during 5 July—22 August, and the 460,000 acre feet of stored water from the USBR projects and Brownlee Reservoir was released as observed. See Connor (2003) and Yearsley (2001) for more detailed methods.

The flow and temperature exposure indices were recalculated for each of the cohorts in 2004 by use of the approximated flows and simulated temperatures under the baseline and July-August scenarios. The differences between the joint probability of active migration and survival predicted for each cohort under the observed and baseline scenarios were used to determine if flow augmentation increased survival of wild fall Chinook salmon subyearlings of Snake River origin in 2004. The differences between this joint probability predicted for each cohort under the observed and July-August scenarios were used to determine if saving Dworshak Reservoir water for release in September decreased the joint probability of active migration and survival of wild fall Chinook salmon subyearlings of Snake River origin in 2004.

Results and Discussion

Accuracy of Origin Classification

In 2004, we recaptured 623 previously PIT-tagged fish. Within and across-origin classification accuracy were 100% (Table 1).

Early Life History and Growth

In 2004, a total of 28,847 wild fall Chinook salmon was captured by beach seine (including marked fish captured more than once). Of these; 10,416 were fry. Fry emergence during 2004 occurred earlier in the upper reach than in the lower reach of the Snake River based on time of fry presence (Table 2). Fry emergence in 2004 in the upper reach was 2-d earlier than the 1995—2003 median of 20 April and 1-d earlier than the 1992—2003 median of 3 May in the lower reach.

Table 1.—Within- and across-origin classification accuracy (%) for distinguishing between wild and hatchery subyearling fall Chinook salmon in 2004. Classification of origin (hatchery vs. wild) was based on a subjective assessment of body morphology, and then verified with PIT-tag records.

Actual origin	*n*	Number classified into each origin		Classification accuracy	
		Hatchery	Wild	Within	Across
Hatchery	6	6	0	100	100
Wild	617	0	617	100	

Table 2.—Emergence timing (given as Sunday's date for each week) of wild fall Chinook salmon fry in the upper and lower reaches of the Snake River, 1992—2004. The range of dates is given in parentheses.

Year	*N*	Median dates of presence
	Snake River upper reach	
1995	117	23 Apr (02 Apr to 21 May)
1996	14	28 Apr (14 Apr to 05 May)
1997	1	20 Apr (N/A)
1998	101	19 Apr (12 Apr to 10 May)
1999	97	02 May (04 Apr to 23 May)
2000	683	09 Apr (02 Apr to 14 May)
2001[a]	552	29 Apr (01 Apr to 20 May)
2002[a]	2,289	21 Apr (31 Mar to 02 Jun)
2003[a]	962	13 Apr (23 Mar to 25 May)
2004[a]	6,120	18 Apr (21 Mar to 23 May)
	Snake River lower reach	
1992	355	26 Apr (29 Mar to 24 May)
1993	199	16 May (04 Apr to 20 Jun)
1994	440	15 May (03 Apr to 05 Jun)
1995	257	30 Apr (02 Apr to 04 Jun)
1996	268	05 May (14 Apr to 23 Jun)
1997	114	04 May (20 Apr to 29 Jun)
1998	322	26 Apr (12 Apr to 14 Jun)
1999	278	02 May (04 Apr to 27 Jun)
2000	415	09 Apr (02 Apr to 04 Jun)
2001	1,268	06 May (01 Apr to 03 Jun)
2002	960	05 May (01 Apr to 16 Jun)
2003	1,726	20 Apr (23 Mar to 22 Jun)
2004	4,296	02 May (21 Mar to 06 Jun)

[a]adjusted by removing fish collected non-randomly at rkm 366.

A total of 18,431 of the wild fall Chinook salmon captured in 2004 were parr. Shoreline rearing by parr during 2004 occurred earlier in the upper reach than in the lower reach of the Snake River (Table 3). Rearing timing in the upper reach was 21 d earlier than the 1995—2003 median of 23 May and was 12 d earlier than the 1992—2003 median of 28 May in the lower reach.

Table 3.—Rearing timing (given as Sunday's date for each week) of wild fall Chinook salmon parr in the upper and lower reaches of the Snake River, 1992—2004. The range of dates is given in parentheses.

Year	*N*	Median dates of presence
	Snake River upper reach	
1995	985	28 May (09 Apr to 18 Jun)
1996	118	12 May (14 Apr to 16 Jun)
1997	119	25 May (20 Apr to 15 Jun)
1998	1,078	17 May (12 Apr to 05 Jul)
1999	1,493	23 May (11 Apr to 27 Jun)
2000	1,064	23 Apr (02 Apr to 11 Jun)
2001[a]	793	29 Apr (01 Apr to 10 Jun)
2002[a]	3,013	29 Apr (07 Apr to 30 Jun)
2003[a]	4,523	04 May (23 Mar to 22 Jun)
2004[a]	6,318	02 May (28 Mar to 20 Jun)
	Snake River lower reach	
1992	1,765	17 May (29 Mar to 07 Jun)
1993	2,215	06 Jun (11 Apr to 18 Jul)
1994	4,346	29 May (03 Apr to 10 Jul)
1995	1,408	04 Jun (02 Apr to 02 Jul)
1996	756	26 May (14 Apr to 14 Jul)
1997	938	08 Jun (20 Apr to 13 Jul)
1998	2,512	31 May (12 Apr to 05 Jul)
1999	1,647	06 Jun (04 Apr to 11 Jul)
2000	1,578	14 May (02 Apr to 25 Jun)
2001	3,078	20 May (01 Apr to 17 Jun)
2002	3,619	26 May (01 Apr to 07 Jul)
2003	6,814	11 May (23 Mar to 29 Jun)
2004	12,113	16 May (28 Mar to 27 Jun)

[a]adjusted by removing fish collected non-randomly at rkm 366.

A total of 5,476 of the wild fall Chinook salmon parr was PIT tagged in 2004. Of these, 572 were recaptured during beach seining. Mean growth rate was slightly higher for parr in the upper reach than in the lower reach of the Snake River (Table 4). Growth rate in 2004 for fish in the upper reach was an average of 0.2 mm/d lower than the 1995—2003 average of 1.2 mm/d, whereas growth rate in the lower reach was equal to the 1992—2003 average of 0.9 mm/d.

Table 4.—Mean growth rates (mm/d±SD) for wild fall Chinook salmon parr collected in the upper and lower reaches of the Snake River, 1992—2004. Sample sizes are given in parentheses

Year	Growth rate by reach	
	Upper reach	Lower reach
1992		0.9±0.1 (66)
1993		0.7±0.4 (202)
1994		1.1±0.4 (341)
1995	1.2±0.3 (148)	1.0±0.4 (78)
1996	1.1±0.3 (19)	0.9±0.4 (49)
1997	1.3±0.3 (20)	0.8±0.3 (80)
1998	1.1±0.3 (112)	0.9±0.3 (129)
1999	1.3±0.3 (171)	1.1±0.3 (92)
2000	1.3±0.2 (90)	1.0±0.3 (40)
2001[a]	1.2±0.1 (12)	0.9±0.2 (123)
2002[a]	1.1±0.2 (170)	0.9±0.2 (186)
2003[a]	1.0±0.3 (360)	0.8±0.2 (186)
2004[a]	1.0±0.3 (150)	0.9±0.2 (422)

[a]adjusted by removing fish collected non-randomly at rkm 366.

A total of 1,783 of the 5,476 parr that was PIT tagged was detected as smolts passed Lower Granite Dam. Passage of smolts from the upper reach was earlier than passage of smolts from the lower reach Table 5). The median date of passage in 2004 for smolts from the upper reach was 11 d earlier than during 1995—2003. The median date of passage in 2004 for smolts from the lower reach was 20 d earlier than during 1992—2003.

Table 5.—Smolt migration timing at Lower Granite Dam for wild fall Chinook salmon that were initially captured, PIT tagged, and released in the upper and lower reaches of the Snake River, 1992—2004. The range of dates is given in parentheses.

Year	*N*	Median dates of detection
	Snake River upper reach	
1995	203	18 Jul (04 Jun to 24 Oct)
1996	19	04 Jul (20 May to 25 Jul)
1997	22	27 Jun (04 Jun to 13 Aug)
1998	173	07 Jul (19 May to 21 Aug)
1999	319	03 Jul (02 Jun to 28 Aug)
2000	72	27 Jun (06 May to 18 Jul)
2001[a]	10	12 Jul (24 Jun to 17 Oct)
2002[a]	95	01 Jul (30 May to 27 Jul)
2003[a]	304	25 Jun (22 May to 6 Aug)
2004[a]	185	23 Jun (20 May to 8 Aug)
	Snake River lower reach	
1992	39	20 Jun (04 May to 21 Jul)
1993	234	21 Jul (31 May to 25 Oct)
1994	193	17 Jul (23 May to 01 Nov)
1995	238	01 Aug (02 Jun to 26 Oct)
1996	126	22 Jul (17 May to 31 Oct)
1997	97	16 Jul (14 Jun to 13 Oct)
1998	380	11 Jul (29 May to 19 Oct)
1999	241	25 Jul (01 Jun to 30 Aug)
2000	257	02 Jul (18 May to 28 Oct)
2001	185	07 Jul (16 May to 26 Oct)
2002	395	06 Jul (06 Jun to 30 Sep)
2003	814	29 Jun (03 May to 19 Oct)
2004	1,598	24 Jun (01 May to 25 Sep)

[a]adjusted by removing fish collected non-randomly at rkm 366.

Passage Forecasting

Combining data collected on fish from the upper and lower Snake River reaches in 2004, showed that 50% of the smolts passed Lower Granite Dam by 24 June and passage was 95% complete by 10 July (Figure 2). The passage forecast provided to managers in 2004 forecasted that 50% of the smolts would pass Lower Granite Dam by 21 (3 d early) June and would be 95% complete by 26 June (14 d early; Figure 2). Mean daily error during July and August was 4.4 $\pm$ 5.0% SD and 0.4 $\pm$ 0.1% SD, respectively. Observed passage was higher than the upper 90% forecast interval for six d in late June (Figure 2).

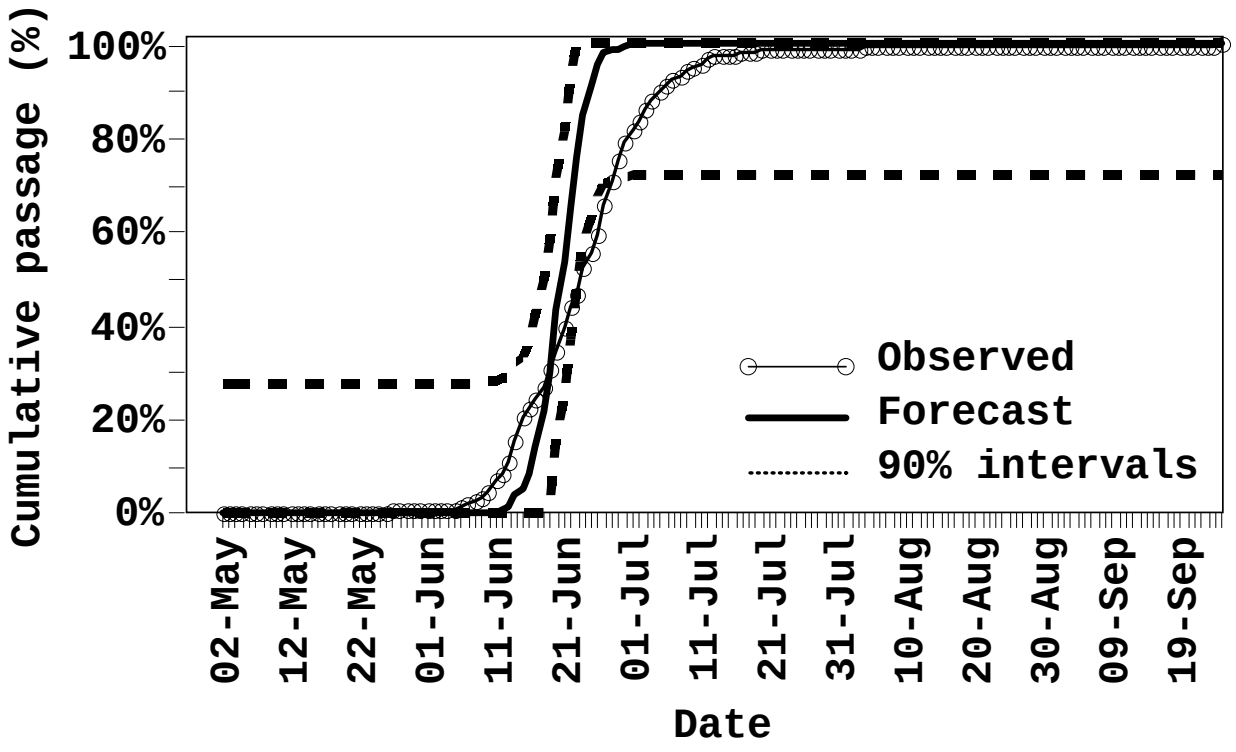


Figure 2.—Forecast of passage at Lower Granite Dam for wild fall Chinook salmon subyearlings implanted with PIT tags in the Snake River in 2004. Fish collected non-randomly at rkm 366 were not included in the observed passage distribution.

Active Migration and Survival

The mean joint probability of active migration and survival to the tailrace of Lower Granite Dam for the four cohorts in 2004 was 44.5% (Table 7). This was 4.5 percentage points lower than the 1998—2003 median of 49.0%.

Table 7.—Estimates of the joint probability of active migration and survival (%±SE) to the tailrace of Lower Granite Dam for cohorts of wild fall Chinook salmon subyearlings, 1998—2004.

	Survival by cohort				Annual
Year	1	2	3	4	means
1998	70.8±2.9	66.1±3.3	52.8±3.1	35.6±2.9	56.3±7.9
1999	87.7±4.6	77.0±3.8	81.2±5.8	36.4±3.5	70.6±11.6
2000	57.1±4.1	53.4±4.2	44.4±3.6	35.7±4.3	47.7±4.8
2001[a]	41.3±3.2	20.4±2.6	16.5±2.9	5.6±2.6	21.0±7.5
2002[a]	55.7±3.0	48.4±3.0	40.5±3.1	18.3±2.0	40.7±8.1
2003[a]	58.7±2.4	52.7±2.2	51.4±2.0	38.2±1.8	50.3±4.3
2004[a]	47.4±1.0	49.6±1.0	46.5±1.0	34.6±1.0	44.5±3.4
All	61.9±6.4	53.0±7.8	47.8±8.6	28.3±5.4	

[a]adjusted by removing fish collected non-randomly at rkm 366.

The Efficacy of Summer Flow Augmentation

The values of the independent variables entered into the product moment correlation analysis are given in Table 8. Of the four independent variables, release date was the only variable that was not significantly correlated with the joint probability of active migration and survival (r = - 0.28, P = 0.15). Fork length (r = 0.64, P = 0.0003) and flow (r = 0.86, $P <$ 0.0001) were directly proportional to this joint probability, whereas temperature was inversely proportional to this joint probability (r = - 0.60, P = 0.0008). These results are consistent with several studies (Kjelson et al. 1982; Kjelson and Brandes 1989; Connor et al. 1998, 2003b, 2004; Smith et al. 2003).

The variables fork length, flow, and temperature as well as all the possible interaction terms were entered into the starting multiple regression model. The final model included temperature, a temperature • fork length interaction term (interaction 1), and a temperature • flow interaction term (interaction 2). The regression equation was: 83.32417 - 10.31009 temperature + 0.08740 interaction 1 + 0.00136 interaction 2. This model had an R^2 value of 0.85 (N = 28, P < 0.0001) and the slope coefficients for temperature and the interaction terms varied significantly ($P < 0.05$) from zero.

The absence of main effect terms for fork length and temperature does not imply that fork and temperature do not effect the joint probability of active migration and surviving, rather that the effects of these two factors are dependent on temperature. Interaction 1 in the model functions to change the prediction of the joint probability at a constant temperature according to fork length, whereas interaction 2 changes the prediction of the probability based on flow. When holding temperature constant at 18°C and flow constant at 2,000 m^3/s, the model predicts a rate of active migration and survival of 41% for a 60-mm fish and 57% for a 70-mm fish. The same types of examples can be calculated for fork length. The interaction terms function to account for the patterns seen in Tables 7 and 8. The biological implications of the interaction terms are as follows. Interaction term 1 implies that predators will consume more small fish than large fish at a given temperature (i.e., size selective predation; e.g., Tabor et al. 1993). Interaction term 2 implies that fish will move seaward more rapidly at a given temperature under a higher flow than under a lower flow (e.g., Connor et. al. 2003a) and the faster moving fish will survive at higher rates than the slower moving fish.

In 2004, observed outflows during the period when the majority of the PIT-tagged smolts passed Lower Granite Dam in 2004 (range, 3 June to 24 July) were higher (mean, 1,545 $\pm$ 700 m^3/s SD) and temperatures (mean, 17.5 $\pm$ 2.9°C) cooler in the tailrace than the approximated outflows (mean, 1,744 $\pm$ 844 m^3/s SD) and simulated temperatures (mean, 18.1 $\pm$ 3.4°C) under the baseline scenario (Figure 3). These results support the conclusion that summer flow augmentation in 2004 increased flow and decreased temperature in Lower Granite Reservoir consistent with previous analyses (Connor et al. 1998; Connor et al. 2003b).

In 2004, observed outflows during the period when the majority of the smolts passed Lower Granite Dam (range, 3 June to 24 July) were lower (mean, 1,545 $\pm$ 700 m^3/s SD) and temperatures (mean, 17.5 $\pm$ 2.9°C) warmer in the tailrace of Lower Granite Dam than the

Table 8.—Predictor variables entered into a product moment correlation analyses as a step to fitting a multiple regression model for predicting the joint probability of wild subyearling fall Chinook salmon actively migrating and surviving to the tailrace of Lower Granite Dam, 1998—2004. Abbreviations: Date = median day of year of release; FL = mean fork length (mm) at release; Flow = a flow (m^3/s) exposure index calculated as the mean outflow measured at Lower Granite Dam during the period when the majority of smolts passed the dam; and, Degrees = a water temperature (°C) exposure index calculated as the mean temperature measured in the tailrace of Lower Granite Dam during the period when the majority of smolts passed the dam. See Connor et al. (2003b) for exposure index calculations.

Year	Cohort	*N*	Date	FL	Flow	Degrees
1998	1	515	140	80	2,344	17.6
	2	515	141	75	2,021	18.7
	3	515	153	73	1,898	19.0
	4	515	167	70	1,299	19.8
1999	1	441	147	80	2,378	16.3
	2	440	153	77	1,963	17.1
	3	440	152	70	2,116	16.7
	4	440	167	68	1,353	18.3
2000	1	303	130	77	1,510	16.7
	2	302	144	77	1,296	17.6
	3	302	146	77	1,274	17.8
	4	302	158	71	859	18.5
2001[a]	1	340	136	74	761	18.5
	2	340	142	69	743	18.9
	3	340	143	68	753	19.2
	4	341	151	66	745	18.7
2002[a]	1	594	141	74	1,712	16.7
	2	594	143	71	1,620	16.8
	3	594	149	68	1,352	17.4
	4	595	163	69	946	18.7
2003[a]	1	1,164	130	70	2,271	15.7
	2	1,164	136	67	2,221	16.1
	3	1,164	142	67	1,584	17.2
	4	1,166	153	66	1,297	17.9
2004[a]	1	1,369	132	72	1,898	15.8
	2	1,369	139	69	1602	16.8
	3	1,369	140	67	1437	17.1
	4	1,369	153	66	1218	18.1

[a]adjusted by removing fish collected at a non-typical site.

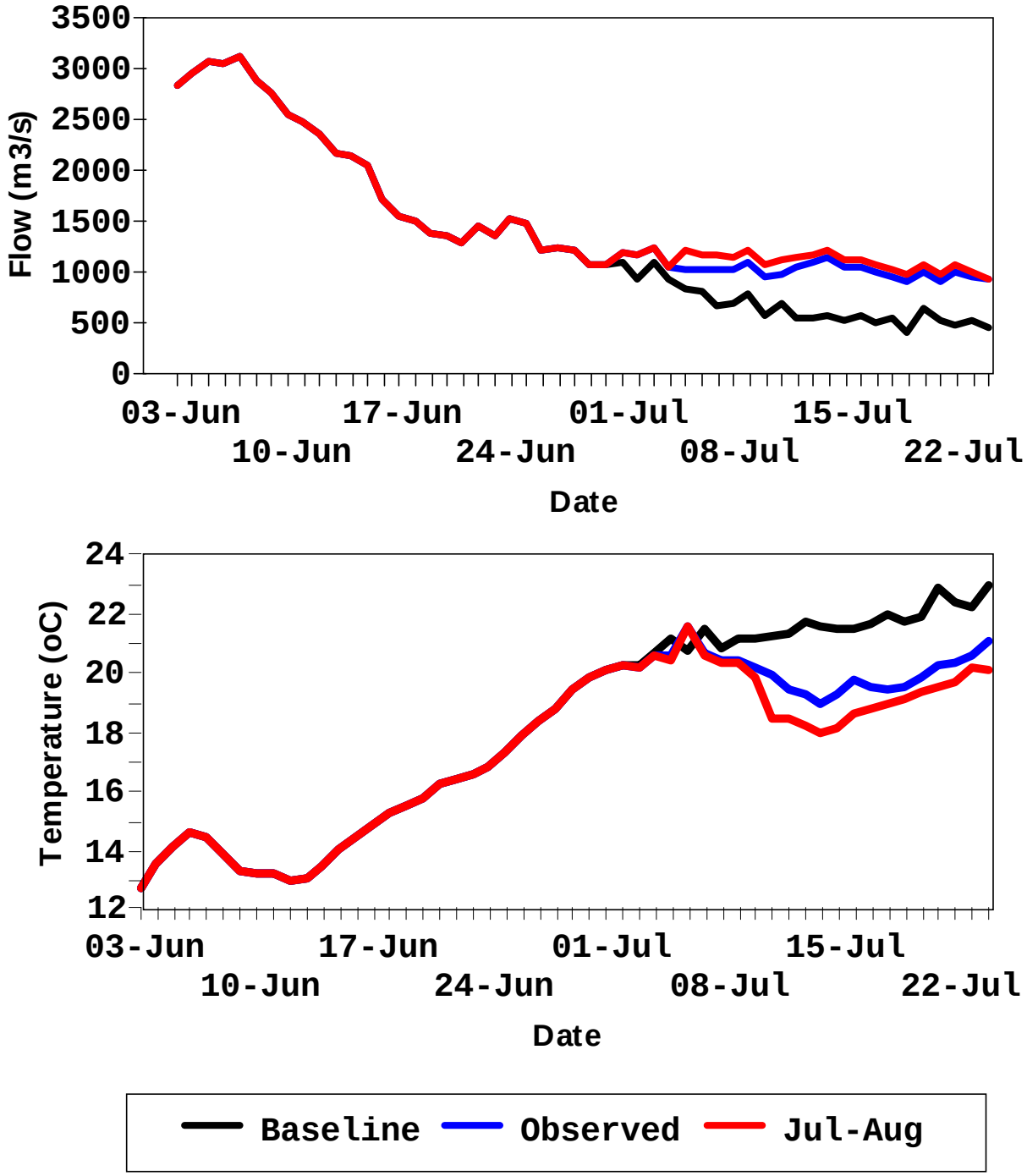


Figure 3.—Daily mean outflows (m^3/s; top panel) and temperatures (°C; bottom panel) in the tailrace of Lower Granite Dam that were observed in 2004 as well as approximated outflows and simulated temperatures under the baseline (i.e., no flow augmentation)and July-August (i.e., no water saved for release in September) scenarios.

approximated outflows (mean, 1,580 ± 675 m^3/s SD) and simulated temperatures (mean, 17.3 ± 2.9 °C) under the July-August scenario (Figure 3).

The predicted joint probability of active migration and survival for cohorts 1, 2, and 3 differed by only 1—3 percentage points when between the baseline and observed scenarios (Table 9). The predicted joint probability for cohort 4 was 8% points higher under the observed scenario than under the baseline scenario (Table 9). These modeling results indicate that summer flow augmentation increased the joint probability of active migration and survival of the latest migrating wild fall Chinook salmon subyearlings of Snake River origin in 2004.

The differences in the predicted joint probability of active migration and survival among all four cohorts under the observed and July-August scenarios ranged from only 0-2 percentage points (Table 9). These modeling results indicate that saving water during July and August for release in September to protect wild Clearwater River subyearlings did not have a large negative effect on active migration and survival of wild fall Chinook salmon subyearlings of Snake River origin in 2004.

Table 9.—The predicted probability of active migration and survival to the tailrace of Lower Granite Dam (Pred ± 95% CLs) for cohorts of wild fall Chinook salmon subyearlings implanted with PIT tags and released at point of capture in the Snake River, 2004. Actual outflows and tailrace temperatures at Lower Granite Dam were used to predict the joint probability under the observed scenario, whereas approximated flows and simulated temperatures were used under the baseline and July-August scenarios. The flow and temperature indices used in the predictions were calculated as described by Connor et al. (2003b).

	Baseline			Observed			July-August		
Cohort	Lower	Pred	Upper	Lower	Pred	Upper	Lower	Pred	Upper
1	54	60	67	54	61	67	54	61	67
2	41	45	49	43	47	51	44	48	52
3	33	37	41	35	40	44	36	41	45
4	15	22	29	24	30	35	26	32	37

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CHAPTER TWO

Temperature Selection by Subyearling Fall Chinook Salmon in Lower Granite Reservoir in Relation to Summer Flow Augmentation in the Snake River

by

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Abstract.—In 2004, we used a combination of temperature monitoring and radio telemetry to evaluate the effects of summer flow augmentation on subyearling fall Chinook salmon in the Snake River. We tagged a total of 99 fish during three release periods with temperature-sensing radio transmitters and monitored their movement in the upper portion of Lower Granite Reservoir using fixed telemetry stations and boat mobile tracking. Mean migration rates through various reaches ranged from 8 to 51 km/d. Radio-tagged juvenile fall Chinook salmon selected a wide range of available water temperatures. The range of temperatures selected was greatest in the confluence area and decreased slightly in a downstream direction as warm and cool water gradually mixed. During the day, most fish used 15-17°C temperatures, but at night they were distributed over 15-20°C temperatures. Downstream of Silcott Island, fish used 16-22°C temperatures but the number of fish detected was low. Overall, most fish occupied water temperatures ranging from 17 to 18°C as they migrated downstream. However, in the first 5 km after release, 28% of all fish were in water >20°C, but in the other downstream reaches only 13-19% of all fish used water >20°C. The only exception was in the 5-km reach below Steptoe in which no fish used water that was warmer than 20°C. Fish passing Red Wolf Bridge used depths ranging from near-surface to 7 m during both the day and night and fish passing Steptoe used depths down to 8 m. Our temperature monitoring efforts showed that cool water releases from Dworshak Reservoir established multiple temperature strata in Lower Granite Reservoir. Differences between surface and bottom temperatures at sample locations were approximately 8°C. Our results suggest that flow augmentation establishes cool water refugia within Lower Granite Reservoir that is selected for by outmigrating subyearling fall Chinook salmon.

Introduction

Fall Chinook salmon in the Snake River display two juvenile life history types—ocean-type and reservoir-type (Connor et al. 2005). Fall Chinook salmon with an ocean-type life history emerge in the spring and migrate seaward in their first year of life as subyearlings. Reservoir-type juveniles delay their seaward migration, overwinter in reservoirs, and migrate seaward in the winter or the following spring as yearlings. Because the outmigration of ocean-type fall Chinook salmon in the Snake River occurs primarily during the summer, flows are typically low and water temperatures are high, sometimes near the upper incipient lethal maximum in surface waters (Tiffan et al. 2003). Consequently, the survival of cohorts of late migrants is often low (Connor 2003).

One of the measures currently implemented to improve summer migratory conditions and survival of ocean-type subyearlings is the release of additional water from upstream storage reservoirs, which is referred to as flow augmentation. The underlying assumption of flow augmentation is that increasing flows will decrease fish travel time, thereby decreasing exposure to predators, disease, and unfavorable environmental conditions, which will subsequently increase survival. Additionally, water released from Dworshak Reservoir in Idaho has a cooling effect on Lower Granite Reservoir, the first reservoir encountered by fish during their seaward migration. Therefore, Dworshak releases are believed to increase survival by reducing exposure to high water temperatures. However, one concern that has been raised is whether cool water from the Clearwater River delays the migration of subyearlings at the confluence of the Snake and Clearwater rivers. Furthermore, for the cooling benefit of flow augmentation from Dworshak Reservoir to be realized, fish must be able to detect and use this cooler water that is typically in the lower portion of the water column (Cook et al 2006). In this study, we attempted to determine if, and how, migrating subyearlings select cool water resulting from flow augmentation at, and below, the confluence of the Snake and Clearwater rivers.

Methods

In this study, we used radio telemetry to characterize the water temperatures selected by tagged subyearlings as they moved downstream from the confluence for a distance of about 16 km. We used a combination of fixed detections sites and intensive mobile tracking to determine thermal and migration histories of tagged fish. We deployed arrays of thermographs to collect time-series temperature data at selected fixed detection sites, and also collected vertical water temperature profiles at fish detection sited during mobile tracking.

Radio telemetry

Fish collection, tagging, and release.—Subyearling fall Chinook salmon were collected and fitted with radio transmitters (Table 1) at the Lower Granite Dam juvenile fish collection facility (river kilometer (rkm) 173). Tagging occurred from July 26 to August 3 (Release 1), August 9 to August 16 (Release 2) and August 23 to August 31, 2004 Release 3). Radio transmitters (Advanced Telemetry Systems, Inc., Isanti, Minnesota) were surgically implanted according to methods described by Adams et al. (1998). Transmitters were 18 mm long, 7 mm in diameter,

weighed 1.3 g in air, and had a life span of 4 d. Tag accuracy was ±0.5°C. The transmitters operated on unique frequencies and had pulse intervals (in milliseconds) that varied directly with temperature. Of the fish that we tagged, the ratio of tag weight to fish weight did not exceed 5%.

Groups of 3-5 fish were tagged daily for 8 d during each release period. Tagged fish were held for 24-36 h after tagging to monitor short-term delayed mortality and were then transported to the release site near the confluence of the Snake and Clearwater rivers. Fish were transported in 125-L insulated containers with a maximum of 3 fish per container. Fish were released on the Snake River (rkm 224.6), immediately upstream of its confluence with the Clearwater River (Figure 1). Releases of 2-3 fish were made twice daily during the morning (0900-1100 hrs) and evening (1900-2100 hrs).

Table 1.—Release dates, number of fish (N) and mean fork lengths and weights of radio-tagged juvenile fall Chinook salmon released into Lower Granite Reservoir in 2004. Numbers in parentheses represent standard deviations of mean lengths and weights.

Release dates	N	Length (mm)	Weight (g)
July 27 - August 3	32	28.3 (3.7)	130.4 (5.3)
August 10 – August 17	32	30.4 (9.9)	137.4 (4.8)
August 24 – August 31	35	41.4 (5.2)	148.6 (6.0)

Fixed-site monitoring.—The movement of radio-tagged fish was monitored by arrays of antennas and receivers located at various fixed sites in the upper portion of Lower Granite Reservoir (Figure 1). Fixed-sites were established at the U.S. Army Corps of Engineers maintenance yard in Clarkston (rkm 224.6), Clarkston Terminal (rkm 222.7), Red Wolf Bridge (rkm 221.1), Port of Wilma (rkm 218.2), Silcott Island (rkm 213), Steptoe Canyon (rkm 206.5), and Blyton (rkm 194.4). At Steptoe Canyon and Blyton, fixed-sites were established on both sides of the river. We refer to the areas between detection sites as reaches. Four- or nine-element Yagi antennas were used at each site in conjunction with a Lotek SRX receiver, solar panel, and 12-V battery.

Individual transmitters operated on unique frequencies, which limited the total number of fish that could be feasibly monitored without increasing the likelihood of missing fish due to increased scan times. As a result, telemetry receivers located at fixed sites from Silcott Island upstream were downloaded daily and updated with frequencies for fish being released on that particular day. This protocol was only altered when a daily download of an upstream receiver suggested that a fish released during a previous day was still in the area and actively moving. Fixed-site receivers at Steptoe Canyon and Blyton received all frequencies for a given release period prior to the initiation of that release period and downloaded 2-3 times per period to monitor fish exiting the downstream end of the coverage area.

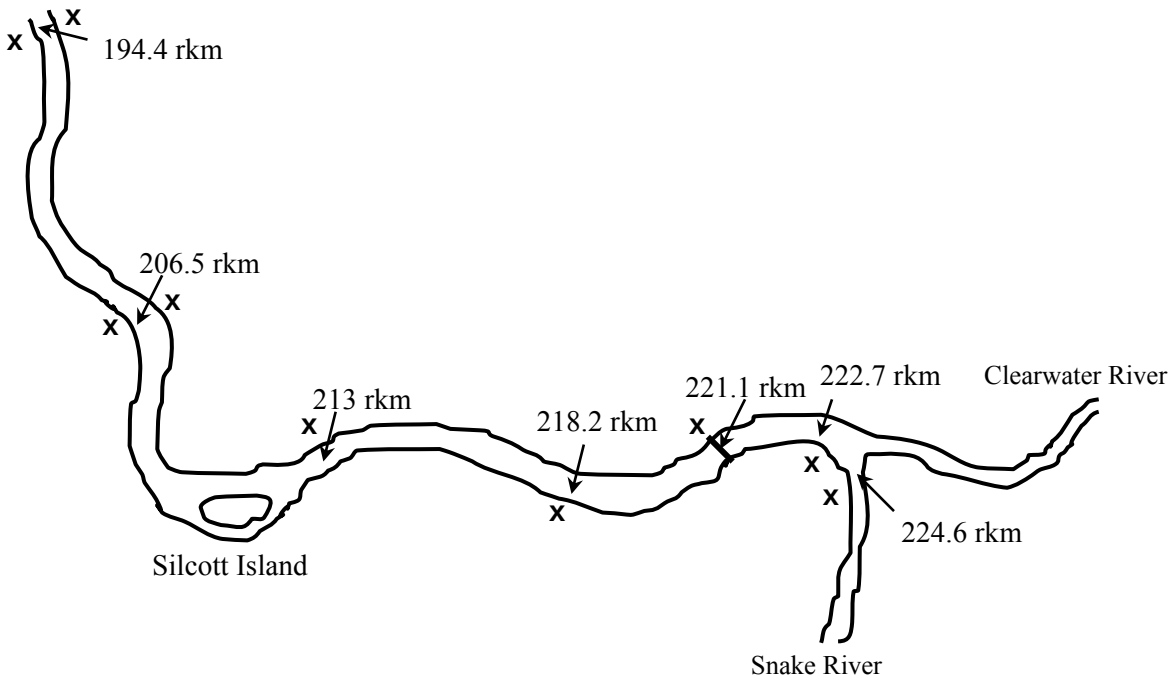


Figure 1.—Map of approximate locations of radio telemetry detection sites used to monitor migrating juvenile fall Chinook salmon from July to August 2004.

Mobile tracking.—Boat mobile tracking was the primary method used to obtain detailed thermal history data on individual fish tagged with temperature-sensing transmitters. Up to 20 h of mobile tracking effort was expended each day to obtain data on fish during both the daytime and nighttime. Mobile tracking crews monitored individual fish continuously with a Lotek SRX receiver set to scan and log temperature records every 3s. Once a fish moved below Silcott Island, tracking crews returned to the confluence area to locate a new fish to monitor. Each hour, fish locations were recorded with a GPS and a vertical temperature profile of the water column was collected with a bathythermograph (BT). The bathythermograph recorded temperature and depth every 0.3 s.

Temperature monitoring

Thermograph strings were used to monitor the vertical distribution of water temperatures at selected sites in the Snake and Clearwater rivers. Five to six data-logging thermographs (HOBO Water Temp Pro loggers (model H20-001), Onset Corporation) were deployed at different depths from the surface to the bottom of the river at the Blue Interstate Bridge across the Snake River in Clarkston, railroad bridge across the Clearwater River in Lewiston, ID, Red Wolf Bridge across the Snake River, Steptoe Canyon and forebay of Lower Granite Dam (Table 2). Thermographs were deployed in early February and were retrieved in early September. Temperatures were recorded every 30 min.

Table 2.—Locations and depths of thermograph strings deployed in the Snake and Clearwater rivers in 2004.

Thermograph string location	Depths of individual thermographs (m)
Blue Bridge	0.5, 1.5, 3, 5, 7, 9
Railroad Bridge	0.5, 1.5, 3, 5, 7, 9
Red Wolf Bridge	0.5, 1.5, 3, 5, 7
Steptoe Canyon	0.5, 1.5, 3, 5, 10, 15
Lower Granite Forebay	0.5, 1.5, 3, 5, 10, 15

A BT was used to collect vertical water temperature profiles in the coverage area of each radio-telemetry fixed site. Cross-sectional transects were established and BT profiles were collected at eight points spaced evenly along each transect. The BT recorded time, depth, and temperature every 0.3 s while being lowered and raised to the surface at a rate of 0.5 m/s with an electric winch. The location of each profile was recorded with a GPS. At each profile location, surface temperatures were obtained with a mercury thermometer to check temperatures obtained with the BT.

Data analysis

Data downloaded from radio telemetry receivers were incorporated into a SAS (Statistical Analysis Software; SAS Institute Inc. 2000) database and automatically proofed to remove erroneous data records. Data were also manually proofed to ensure data accuracy and confirm the integrity of questionable data records. Fish detection records were analyzed to examine travel times, migration rates, and temperature exposure. Fish travel times were calculated as the elapsed time between release and first detection at a fixed-detection site, or between first detection at a site and first detection at the next downstream fixed-detection site. Migration rates were calculated as the length of the reach divided by fish travel time in the reach. Individual fish locations determined during mobile tracking were assigned to either daytime (0630 to 1930 hours) or nighttime (1930 to 0630 hours) periods, and moving fish were categorized as either "active" ($\geq$ 0.4 km/h) or "inactive" ($<$ 0.4 km/h). Hourly average temperatures were calculated for each thermograph at each location. River discharge data during our study were obtained from the U.S. Army Corps of Engineers and from the U.S. Geological Survey.

Temperatures selected by juvenile fall Chinook salmon were combined for fixed-site detections and hourly mobile tracking locations for some graphical displays of results. We constructed frequency distributions of temperatures used by fish as they moved downstream during both the day and night. We also calculated the percentage of fish detected in water greater than 20°C within our study area. Because each temperature observation on mobile-tracked fish represents a 3-s interval, the distribution of observations across different temperature intervals represents time of exposure at each temperature. Because individual temperature histories were so variable, we calculated the percentage of time each fish spent at each 1-°C temperature interval to give each fish equal weight. We then determined the median percentage of time fish spent at each temperature interval for each release because data were not normally distributed. Finally, we determined the depth of fish at each hourly location by finding the depth from the BT profile where the fish temperature matched that of the BT. We also determined the depth of fish passing Red Wolf Bridge and Steptoe using our thermograph strings and linear interpolation.

Results

River flows

During the summer of 2004, flows at Lower Granite Dam declined from 115 kcfs in early June to approximately 30 kcfs by mid-July where they remained through the end of August (Figure 2). During this same period, Snake River flows measured at Anatone, Washington (rkm 270) declined from 67 kcfs to approximately 20 kcfs. In 2004, summer flow augmentation from Dworshak Reservior began on July 2 with an outflow of 10 kcfs. From July 12 to August 8, Dworshak Dam outflows increased and stabilized at 12 kcfs, after which outflows were dropped to 10 kcfs and held steady until August 31.

Water temperature

Thermograph strings.—Water temperatures were generally homothermic in the Clearwater River at the Railroad Bridge and ranged from 10 to16°C during our study (Table3; Figure 3). In the Snake River at the Blue Bridge, temperatures were also homothermic and ranged from 20 to 23°C until late July. However, after late July cooler water was present at 7 and 9 m depths and temperatures ranged from 12 to 24°C (Table3; Figure 3). Water temperatures at Red Wolf Bridge (3 km downstream of the confluence) were warmest at the surface (up to 24°C) and decreased incrementally with depth to lows near 15°C (Table3; Figure 4). Similar temperature patterns were observed at Steptoe Canyon, but there temperatures ranged from about 16°C to 22°C (Table3; Figure 4). Water temperatures in the forebay of Lower Granite Dam also decreased incrementally by depth (Figure 5) although minimum water temperatures were relatively high (> 18 ° C). Near-surface water temperatures exceeded 24°C several times throughout the study period. Water temperatures at the greatest depths were generally 6-8° C cooler than surface temperatures at sites monitored by thermograph strings below the confluence of the Snake and Clearwater rivers.

Bathythermograph transects.—Water temperature data collected with a BT indicated that multiple strata layers are present in the Snake River below its confluence with the Snake and Clearwater rivers. Water temperatures at the Blue Bridge are primarily homothermic with the exception of a small amount of cool water located at the bottom near the east shore (Figure 6). At all other transect locations, water temperatures in the lower half of the channel are approximately 4-6° C cooler than surface temperatures (Figures 7-11).

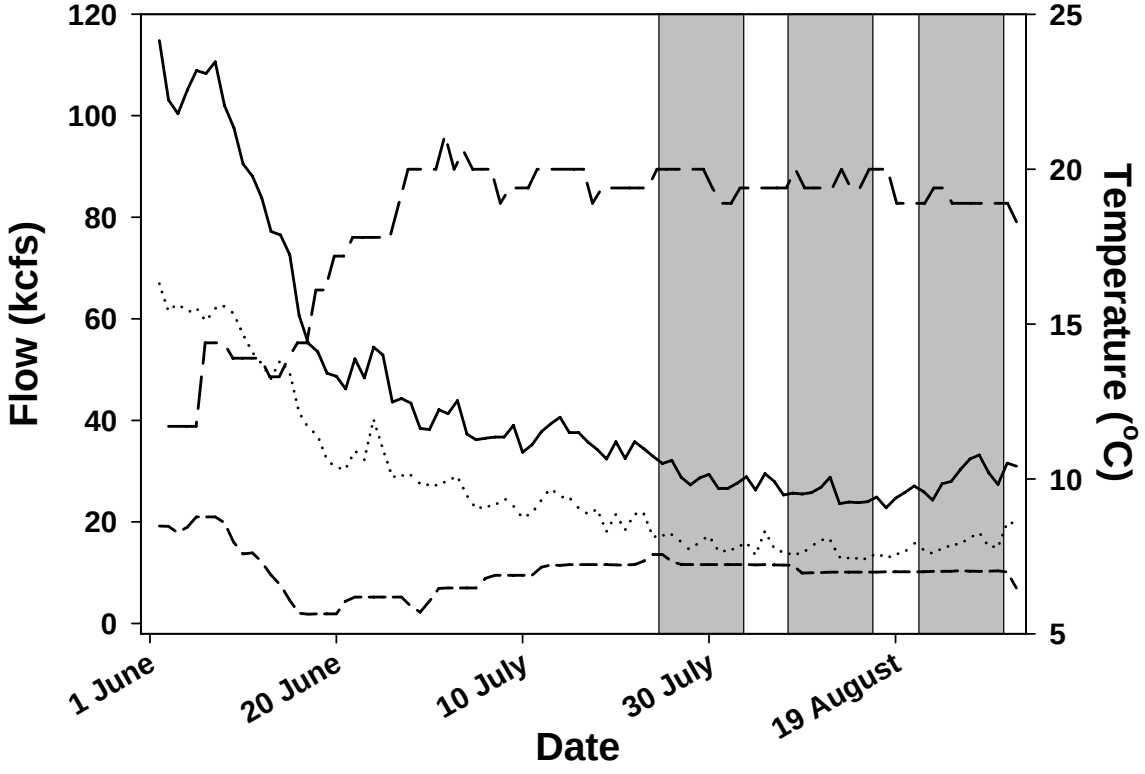


Figure 2.—Snake River flows at Lower Granite Dam (solid line), Anatone, Washington (dotted line), and Clearwater River outflow from Dworshak Dam (short dashed line), and water temperatures measured at the scroll case of Lower Granite Dam (long dashed line), during the summer, 2004. Highlighted areas represent time periods when radio-tagged subyearling fall Chinook salmon were released in Lower Granite Reservoir.

Table 3.—Means and ranges of temperatures measured by thermograph strings deployed in the Snake and Clearwater rivers, July-August 2004.

Location	Depth (m)	Mean Temp (°C)	Temperature Range (min-max)
Blue Bridge	0.5	22.22	19.41 – 24.44
	1.5	22.16	19.38 – 24.47
	3	22.13	19.40 – 24.47
	7	21.29	13.65 – 24.45
	9	18.03	12.41 – 24.37
Railroad Bridge	0.5	13.54	10.35 – 19.05
	1.5	12.77	10.29 – 18.94
	3	12.59	10.11 – 18.75
	5	12.67	10.19 – 18.82
	7	12.74	10.25 – 18.89
	9	12.68	10.20 – 18.82
Red Wolf Bridge	0.5	21.34	18.16 – 24.35
	1.5	20.75	16.61 – 22.81
	3	20.04	14.65 – 22.65
	5	17.78	13.94 – 20.47
	7	16.58	14.06 – 20.05
Steptoe Canyon	0.5	20.19	16.91 – 23.06
	1.0	20.01	16.94 – 22.37
	3	19.76	16.92 – 21.84
	5	19.10	16.36 – 20.96
	10	17.74	15.24 – 20.14
Forebay Barge	0.5	22.80	19.62 – 26.03
	1.5	22.46	19.64 – 25.30
	3	21.96	19.50 – 24.44
	5	21.48	19.27 – 24.28
	10	20.44	18.57 – 23.83
	15	19.96	18.18 – 22.96

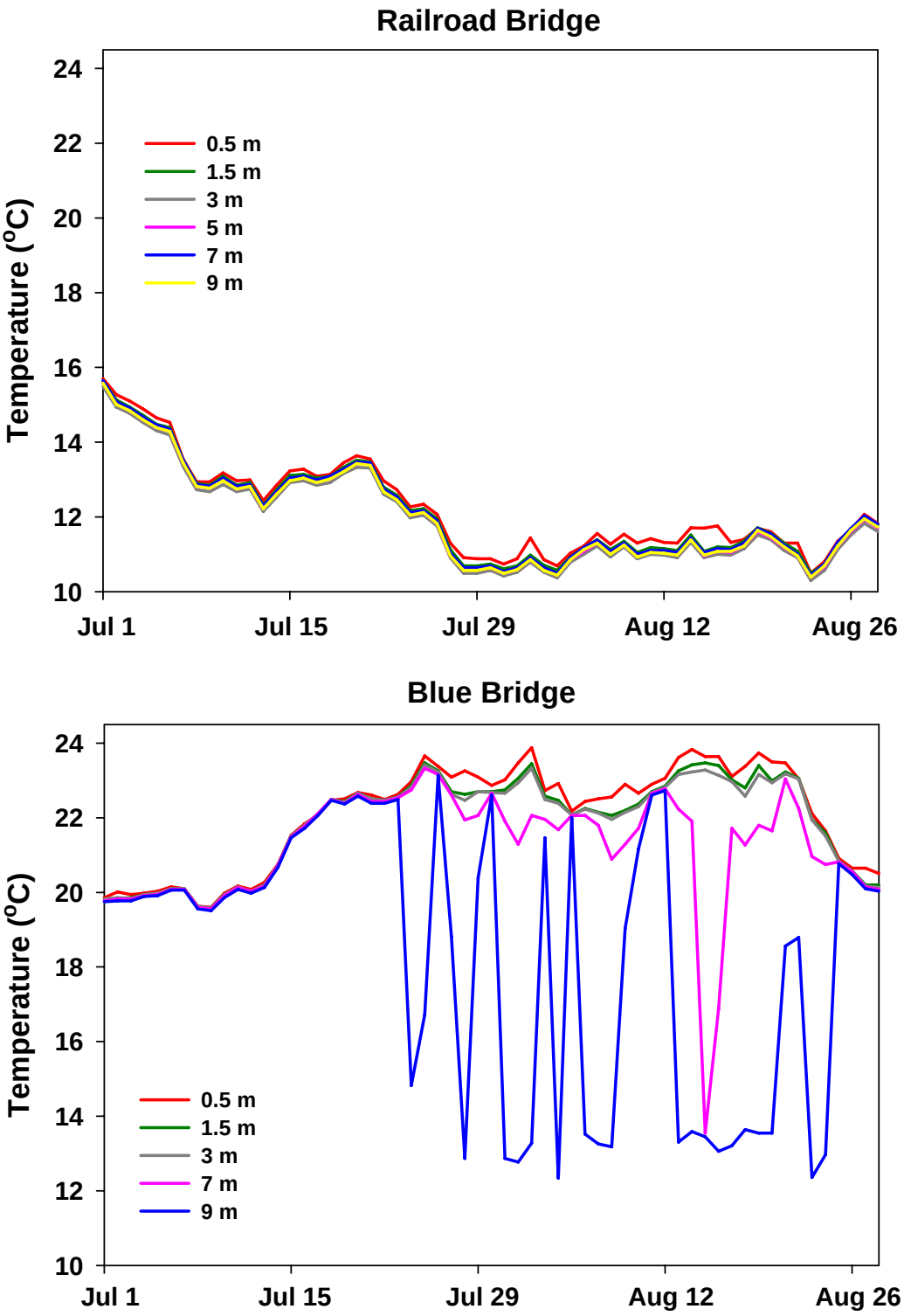


Figure 3.— Temperatures recorded by thermographs deployed at various depths at the Railroad Bridge on the Clearwater River and at the Blue Bridge on the Snake River 2004. Overlap in temperatures may obscure some trend lines.

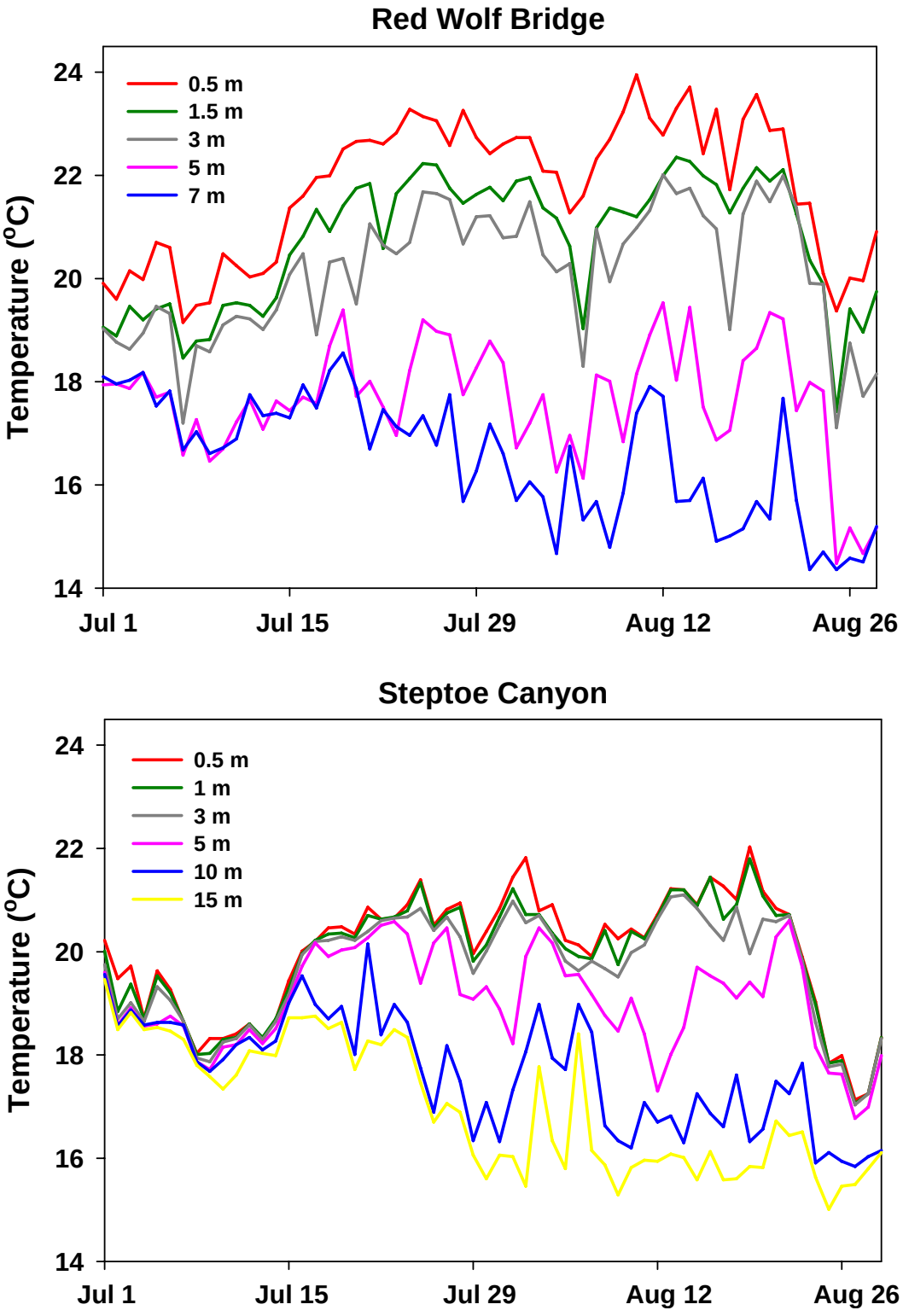


Figure 4.— Temperatures recorded by thermographs deployed at various depths at the Red Wolf Bridge and Steptoe Canyon on the Snake River in 2004. Overlap in temperatures may obscure some trend lines.

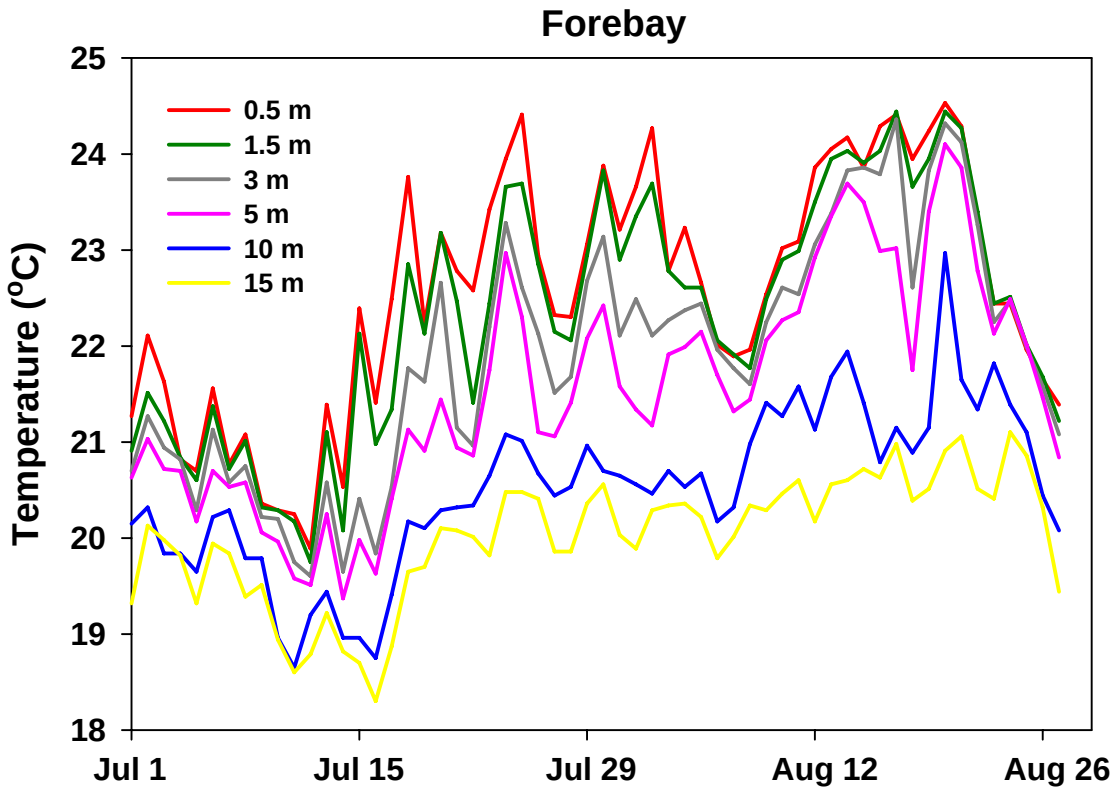


Figure 5.— Temperatures recorded by thermographs deployed at various depths in the forebay of Lower Granite Dam on the Snake River in 2004. Overlap in temperatures may obscure some trend lines.

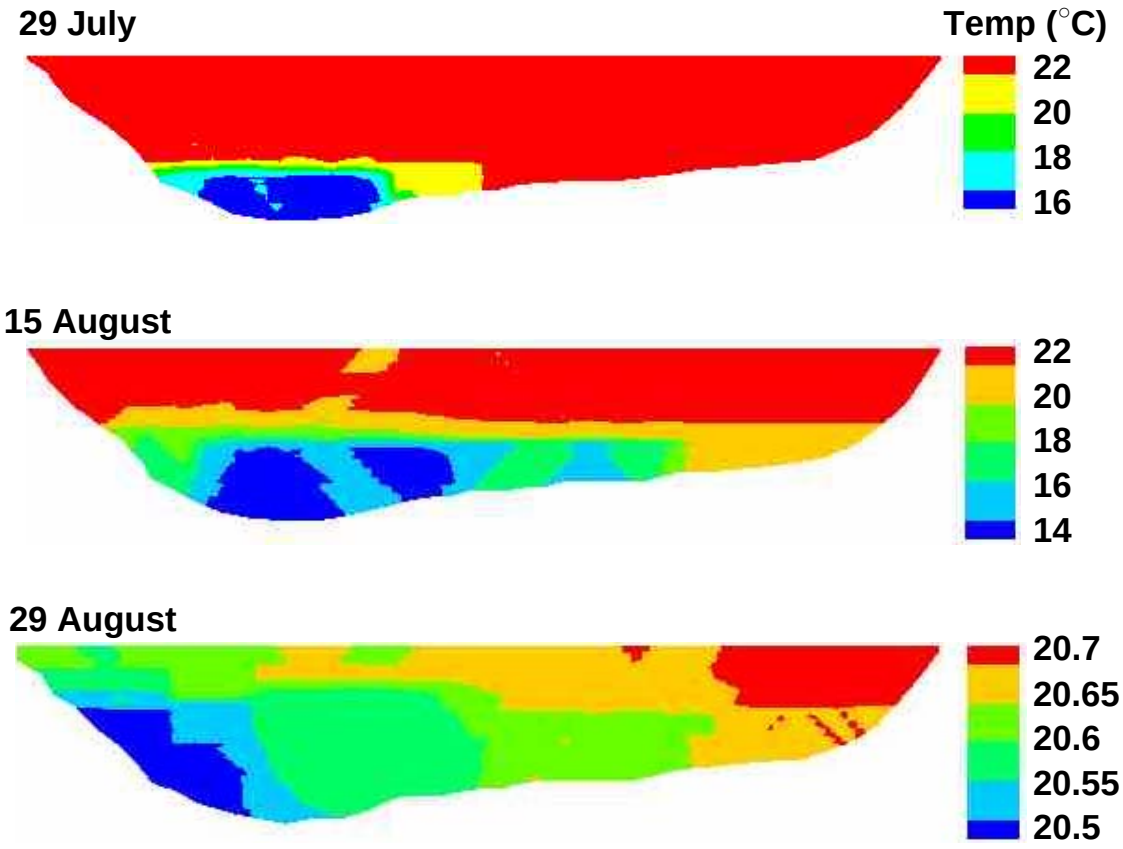


Figure 6.—Graphic representation of temperatures at cross sections at the Blue Bridge on the Snake River in 2004. Vertical temperature profiles were collected with a bathythermograph at points at each cross section. Temperature data were interpolated where data were lacking.

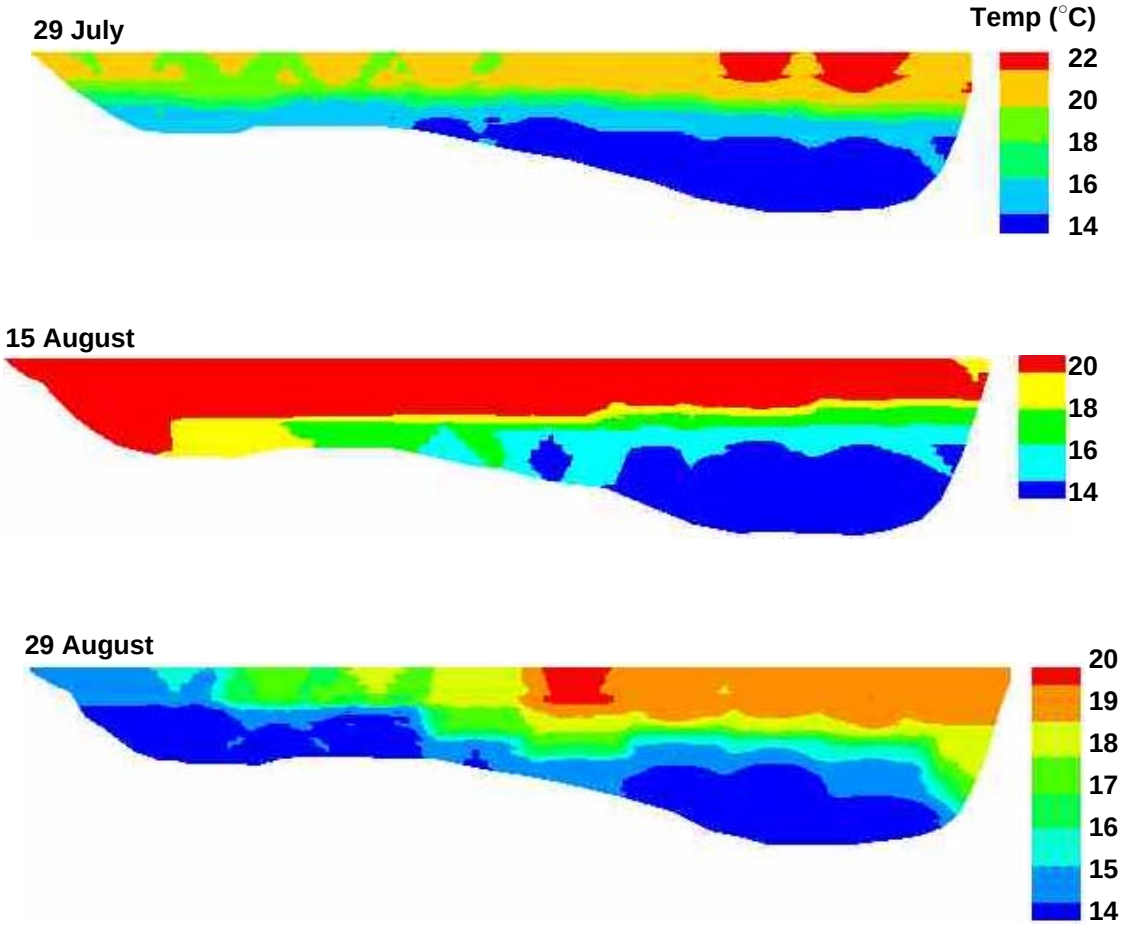


Figure 7.—Graphic representation of temperatures at cross sections at the Clarkston Terminal on the Snake River in 2004. Vertical temperature profiles were collected with a bathythermograph at points at each cross section. Temperature data were interpolated where data were lacking.

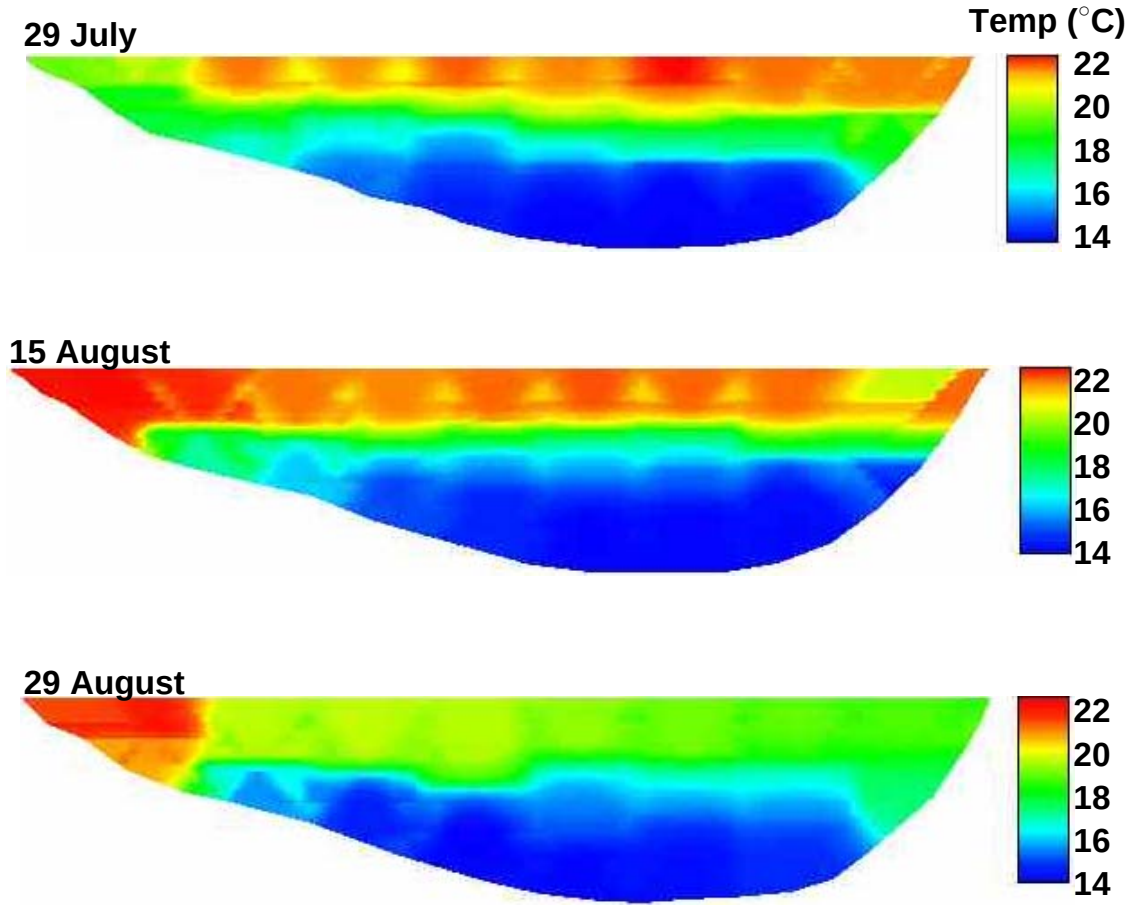


Figure 8.—Graphic representation of temperatures at cross sections at the Red Wolf Bridge on the Snake River in 2004. Vertical temperature profiles were collected with a bathythermograph at points at each cross section. Temperature data were interpolated where data were lacking.

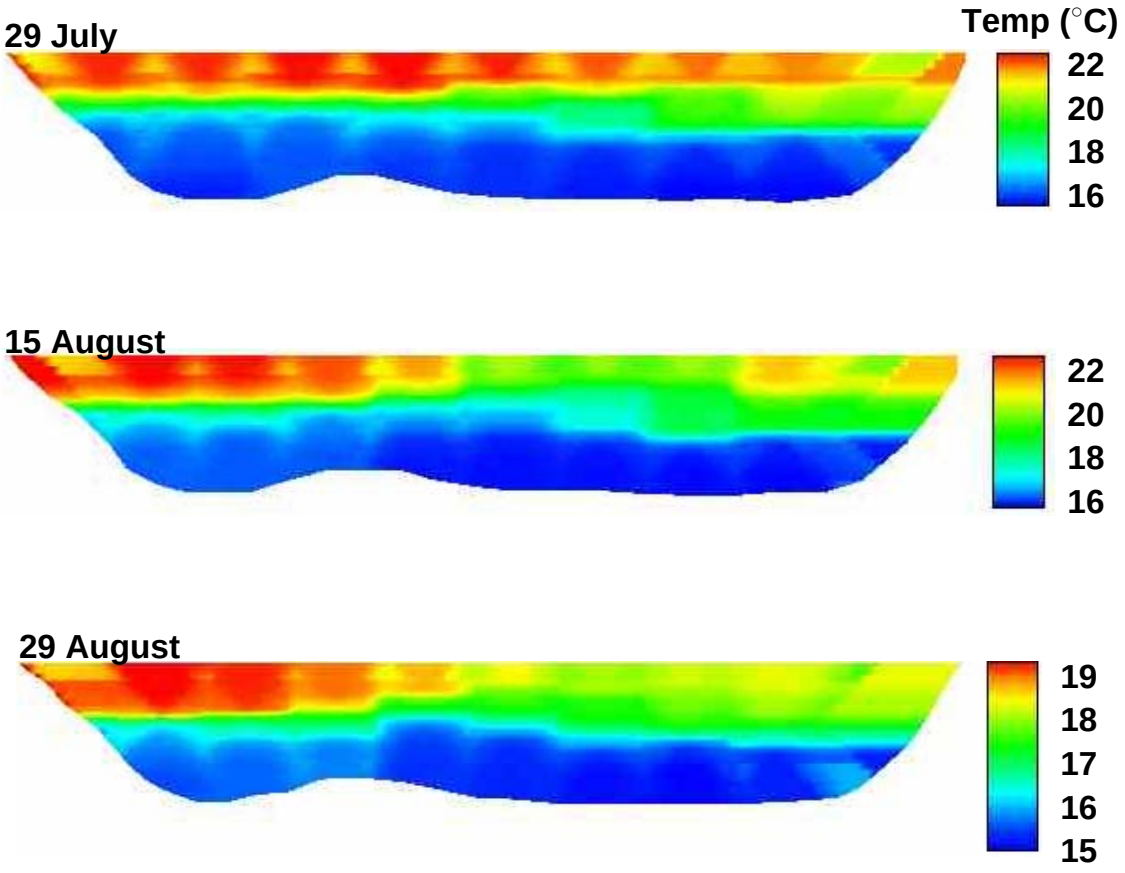


Figure 9.—Graphic representation of temperatures at cross sections at the Port of Wilma on the Snake River in 2004. Vertical temperature profiles were collected with a bathythermograph at points at each cross section. Temperature data were interpolated where data were lacking.

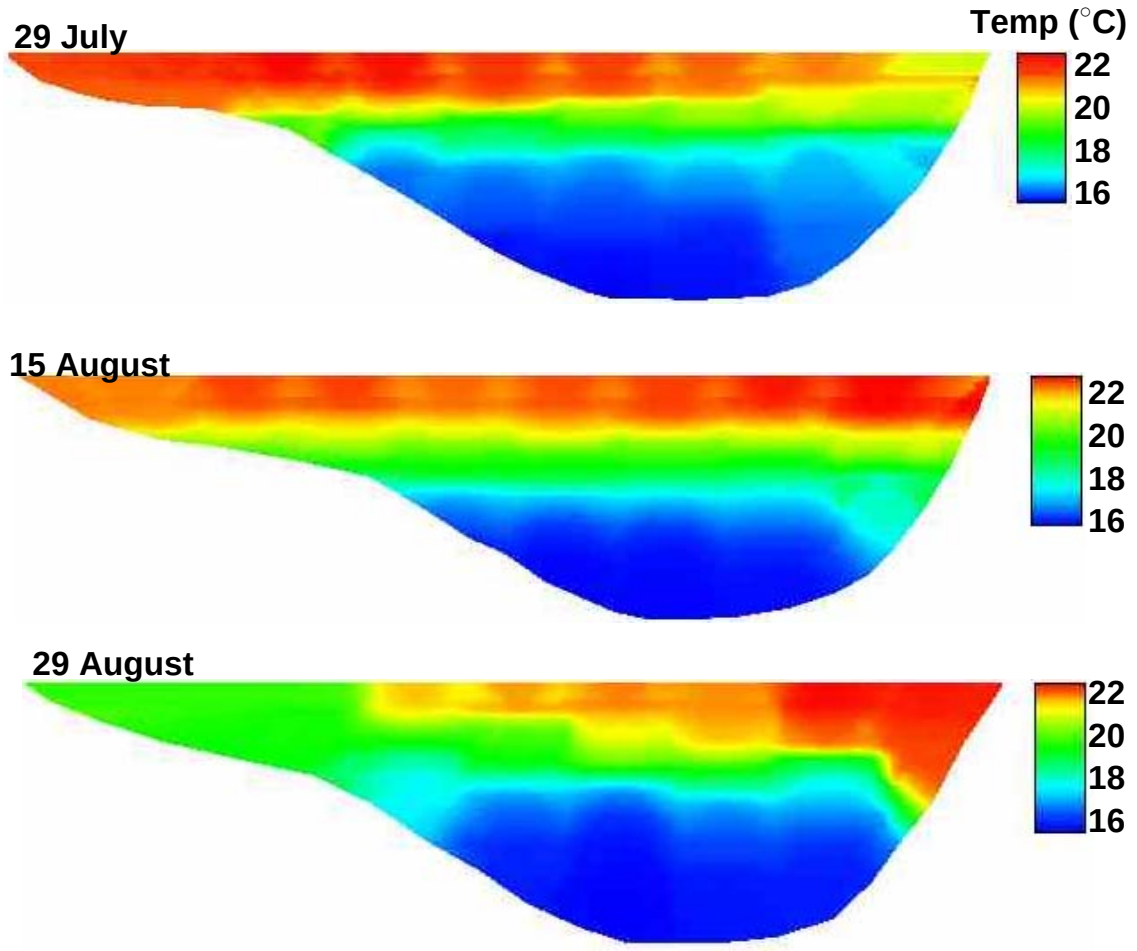


Figure 10.—Graphic representation of temperatures at cross sections at Silcott Island on the Snake River in 2004. Vertical temperature profiles were collected with a bathythermograph at points at each cross section. Temperature data were interpolated where data were lacking.

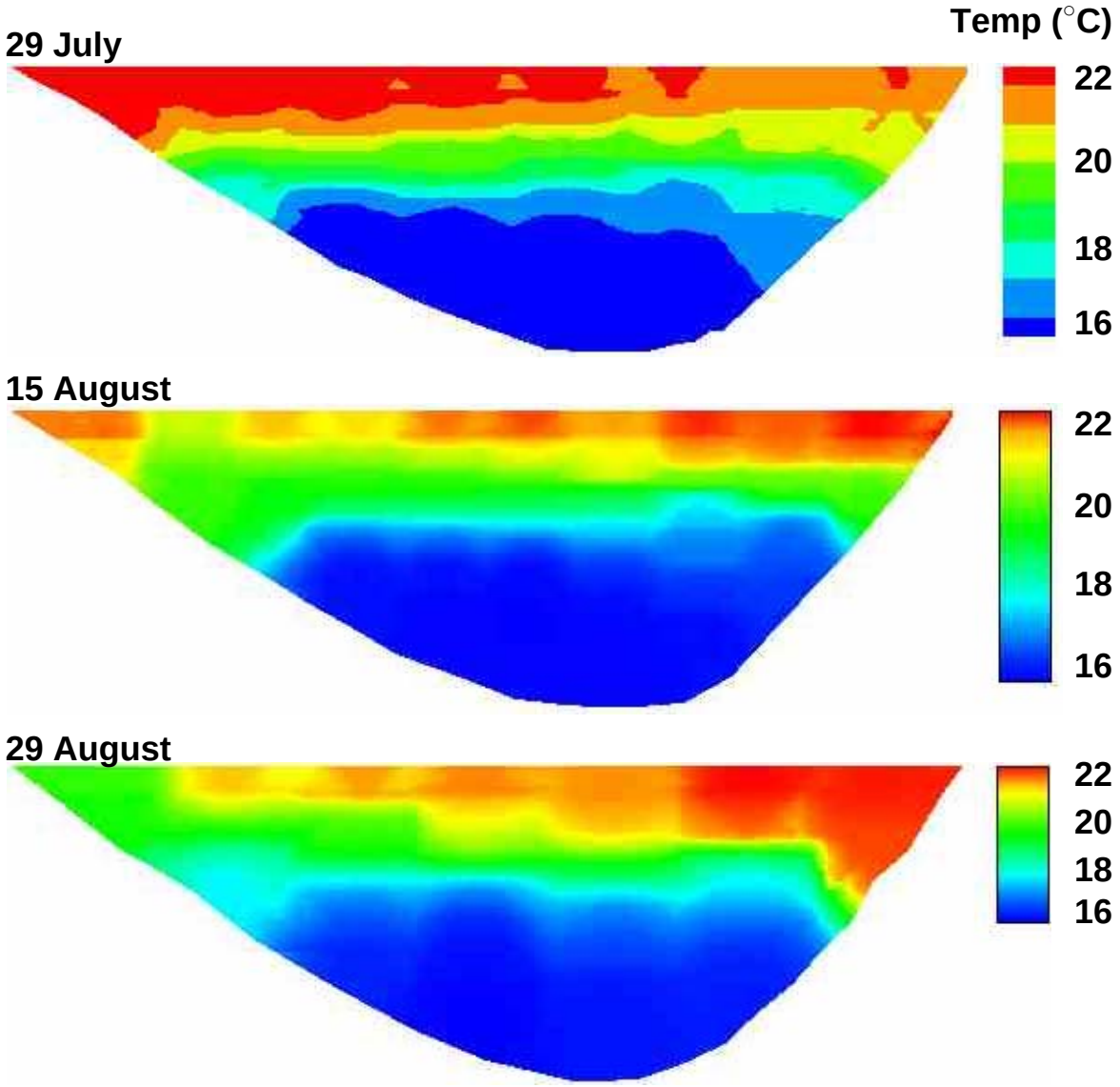


Figure 11.—Graphic representation of temperatures at cross sections at Steptoe Canyon on the Snake River in 2004. Vertical temperature profiles were collected with a bathythermograph at points at each cross section. Temperature data were interpolated where data were lacking.

Radio telemetry

Tagging.—We tagged a total of 99 subyearling fall Chinook salmon with temperature-sensing radio-transmitters in 2004 (Table 1). Mean fork lengths ranged from 125 to 165 mm and mean weights ranged from 24.5 to 53.3 g. We observed no mortality during the 24-36 h post-tagging recovery period in 2004.

Detection efficiency.—We detected 93% (92 of 99) of the subyearling Chinook salmon that were tagged with temperature-sensing radio transmitters in 2004 (Table 4). Detection efficiencies at most sites were generally highest during Release 3 and lowest during Release 2. We detected 87%, 97% and 97% of the fish released during Releases 1, 2 and 3, respectively. The lowest detection efficiency of all release periods occurred during Release 1 at Blyton where only 1 fish (3%) was detected. The highest detection (77%) occurred at the Port of Wilma.

Travel rates.—Travel rates of fall Chinook salmon fitted with radio transmitters varied widely from 1.3 to 117.3 km/d throughout the study period (Figure 12). Mean travel rates among release periods were generally similar, although travel rates during Release 2 were somewhat lower than Releases 1 and 3. Travel rates generally decreased as fish moved downstream with the exception of the Red Wolf Bridge to Port of Wilma reach (rkm 221.1-218.2) where travel increased during all release periods. The observed increase in travel rates during each release period was magnified by sharp decreases in travel rates in the preceding reach (Clarkston Terminal to Red Wolf Bridge; rkm 222.7-221.1).

Mobile tracking.—A total of 38 fish were intensively monitored by mobile tracking crews operating on day and night shifts (Table 5). During daytime tracking, 129 individual hourly locations were recorded. Between these 129 hourly locations, there were 62 (52%) instances when we considered fish to be actively moving and 67 (48%) instances when fish were inactively moving (Table 6). During nighttime tracking, we recorded 234 individual hourly fish locations, between which we observed 167 (71%) instances when fish were actively moving and only 35 (29%) instances when fish were inactive. Mean water temperature for all locations was 17.4° C, and 17.2° C and 17.6° C for day and night locations, respectively.

Temperature selection.— Radio-tagged juvenile fall Chinook salmon selected a wide range of available water temperatures. The range of temperatures selected was greatest in the confluence area and decreased slightly in a downstream direction as warm and cool water gradually mixed (Figure 13). During the day, most fish in the first 5 km downstream of the confluence occupied 16-17°C temperatures, then used 15°C in the next 5 km, and again used 16-17°C farther downstream (Figure 14). During the night, fish were uniformly distributed over 15-20°C temperatures in the first 15 km downstream of the confluence distributed, but then distributions were more variable over a range of 16-22°C farther downstream as the number of detected fish decreased (Figure 14). Overall, most fish occupied water temperatures ranging from 17 to 18°C as they migrated downstream. However, in the first 5 km after release, 28% of all fish were in water >20°C, but in the other downstream reaches only 13-19% of all fish used water >20°C. The only exception was in the 5-km reach below Steptoe in which no fish used water that was warmer than 20°C.

Individual fish experienced different thermal exposures as they migrated downstream (Appendix 1). During Release 1, fish spent the largest amount of time at 20°C, but this result was heavily influenced by two fish that had large numbers of observations at this temperature (Figure 15). After 20°C, the next temperature interval that was most used was 19°C. During Release 2, fish spent the most time at 16°C. During Release 3, fish spent the greatest amount of time at 17°C, however, temperatures were also coolest during this period and the amount of water >20°C was relatively small.

Fish passing Red Wolf Bridge used depths ranging from near-surface to 7 m during both the day and night, and fish passing Steptoe used depths down to 8 m (Figure 16). Fish detected during mobile tracking did not show any seasonal trends in depths used, but fish did use a greater range of depths nearer to the confluence where there was a wider range in temperatures to choose from (Figure 17). Overall, most fish detected during mobile tracking used depths ranging from 3-7 m in our study area (Figure 18).

Table 4.—Detection efficiencies for detection sites used to monitor radio-tagged juvenile fall Chinook salmon in the Snake River, 2004. Numbers in parenthesis denote number of fish detected.

Location	Release 1	Release 2	Release 3	Combined
USACE Yard	41% (13)	34% (11)	66% (23)	47% (47)
Clarkston Terminal	59% (19)	22% (7)	66% (23)	49% (49)
Red Wolf Bridge	41% (13)	31% (10)	57% (20)	43% (43)
Port of Wilma	38% (12)	41% (13)	77% (27)	53% (53)
Silcott Island	44% (14)	25% (8)	69% (24)	46% (46)
Steptoe Canyon	19% (6)	6% (2)	63% (22)	30% (30)
Blyton	3% (1)	19% (6)	57% (20)	27% (27)
Day Tracking Crew	44% (14)	59% (19)	46% (16)	49% (49)
Night Tracking Crew	59% (19)	50% (16)	54% (19)	55% (55)
Overall Detections	84% (27)	97% (31)	97% (34)	93% (92)

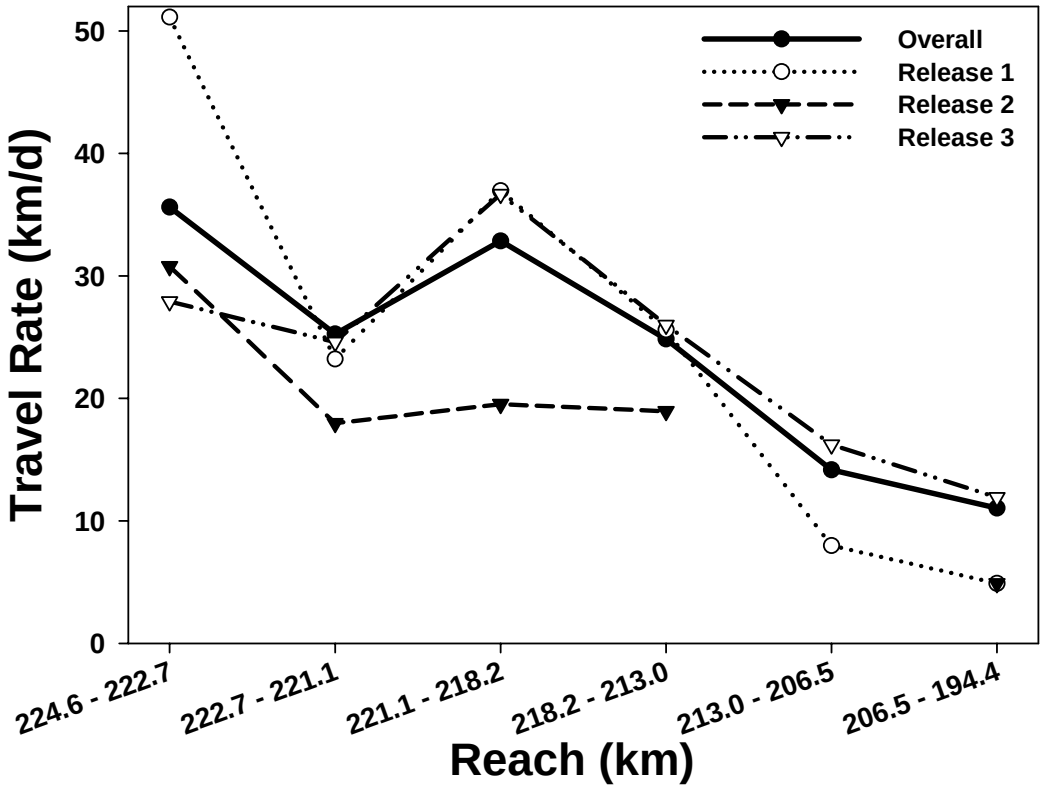


Figure 12.—Mean travel rates of juvenile fall Chinook salmon fitted with temperature-sensing radio transmitters in the Snake River during July and August, 2004. Travel rates are presented for reaches between fixed telemetry detection sites.

Table 5.—Travel rates (range) of juvenile fall Chinook salmon fitted with radio transmitters on the Snake River from July to August, 2004.

Reach	Overall mean travel rate (km/d)	Release 1 mean travel rate (km/d)	Release 2 mean travel rate (km/d)	Release 3 mean travel rate (km/d)
Release to Clarkston Terminal	35.6 (2.3-117.3)	51.1 (6.1-117.3)	30.8 (3.9-66.7)	27.9 (3.3-79.3)
Clarkston Terminal to Red Wolf Bridge	26.3 (1.8–84.4)	29.2 (4.2-84.4)	18.1 (6.3-43.4)	24.6 (1.8-69.5)
Red Wolf Bridge to Port of Wilma	32.8 (4.1–79.1)	36.9 (9.7-79.1)	19.5 (4.1-34.5)	36.7 (7.7-57.3)
Port of Wilma to Silcott Island	24.8 (3.1–61.9)	25.6 (3.1-61.9)	18.9 (5.0-40.0)	26.0 (4.6-47.9)
Silcott Island to Steptoe Canyon	14.2 (1.3–40.1)	8.0 (1.3-12.0)	---	16.2 (7.0-40.1)
Steptoe Canyon to Blyton	11.0 (4.9-31.2)	4.9	4.9	11.9 (5.0-31.2)

Table 6.—Summary of individual locations and activity levels of juvenile fall Chinook salmon fitted with temperature-sensing radio transmitters in the Snake River during July and August 2004. Data was collected by boat mobile tracking crews. Day periods were assigned from 0630 to 1930 hours and all other times were considered night periods. Fish were considered active when they moved greater than 0.4 km between hourly locations.

Period	Activity level	N	Mean water temperature (°C)	Range (min-max)
Day	Active	62	17.5	14.3 – 23.3
Day	Inactive	67	16.9	12.9 – 22.9
Night	Active	167	17.4	13.5 – 21.2
Night	Inactive	35	18.5	14.7 – 23.0

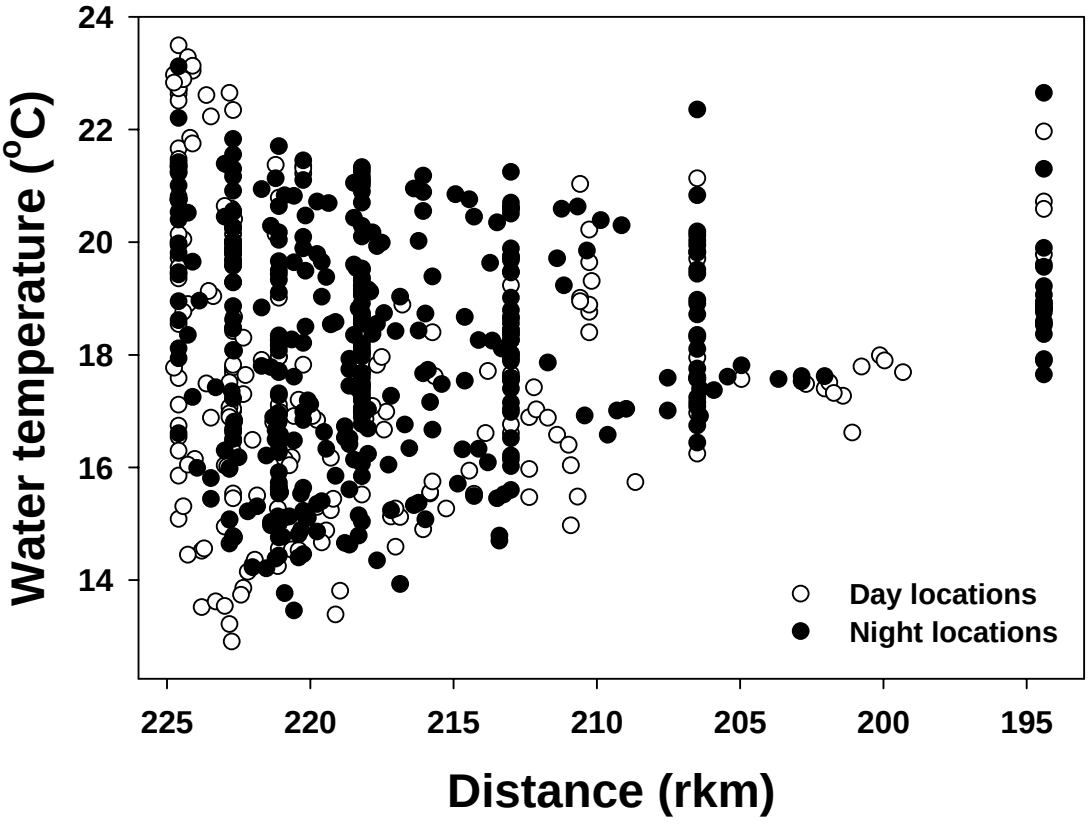


Figure 13.—Water temperatures occupied by juvenile fall Chinook salmon in Lower Granite Reservoir in 2004. Fish were fitted with temperature-sensing radio transmitters and release at the confluence of the Snake and Clearwater rivers. Fish locations were obtained from mobile tracking during the day and the night and from fixed detection sites.

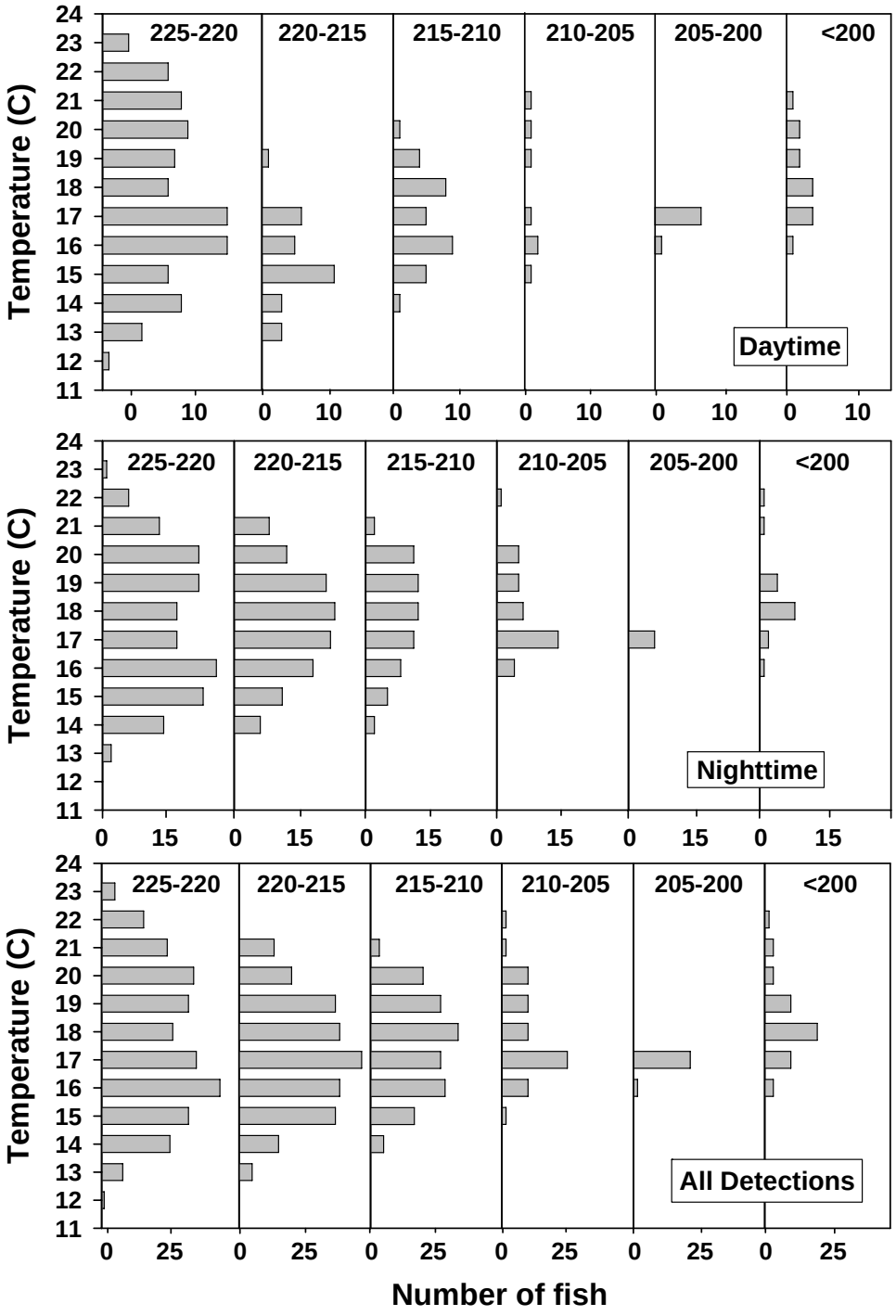


Figure 14.—Distributions of temperatures selected by juvenile fall Chinook salmon in Lower Granite Reservoir in 5-km reaches beginning at the confluence of the Snake and Clearwater rivers (rkm 225). Temperatures were determined from fish were fitted with temperature-sensing radio transmitters and detected at fixed sites and during mobile tracking during the day (top panel) and night (middle panel). Temperatures by location are shown for all fish detected in the bottom panel.

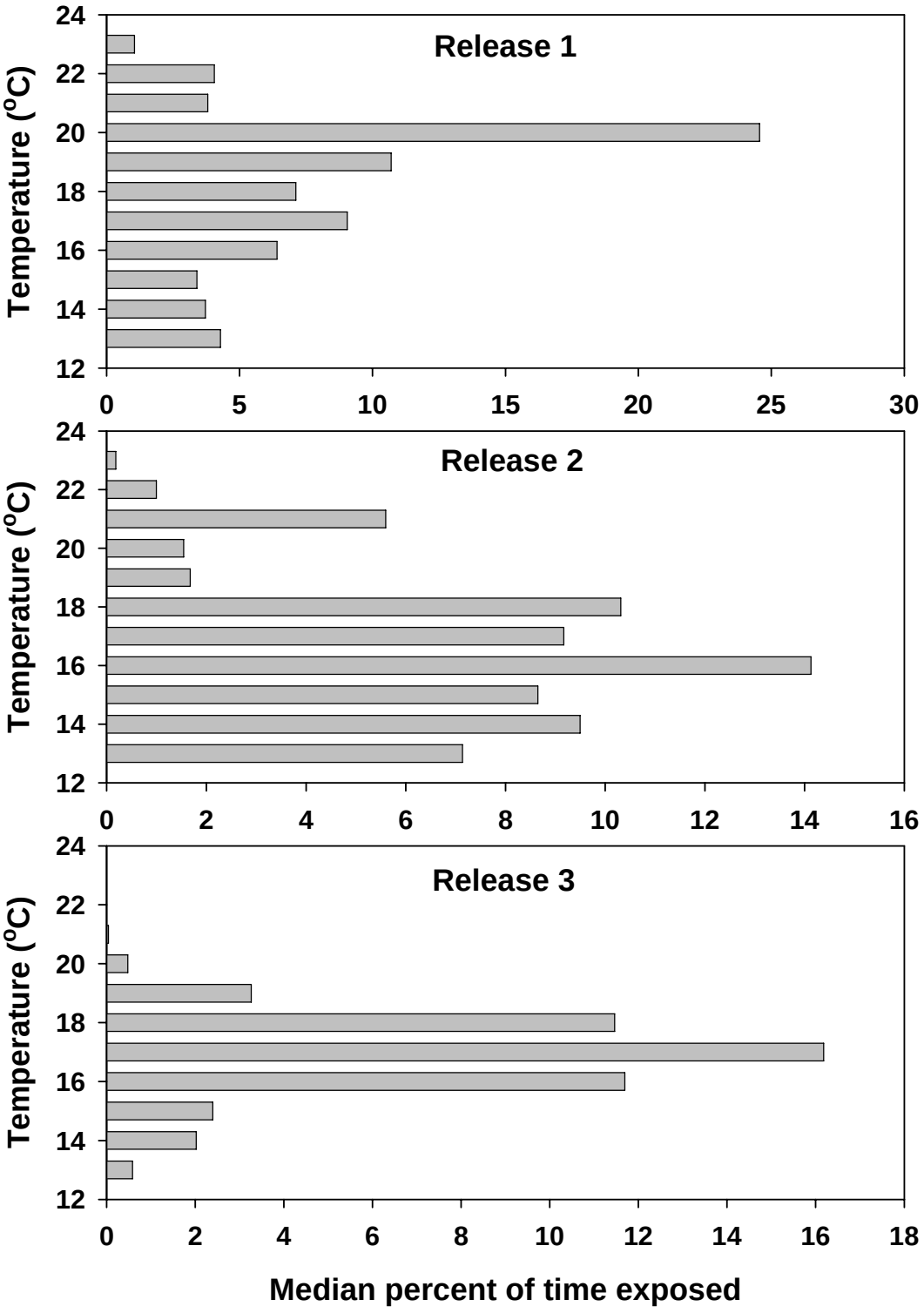


Figure 15.—Median percentage of time that subyearlings were exposed to various water temperatures in the Snake River, 2004. Data were obtained from mobile tracking fish fitted with temperature-sensing radio tags.

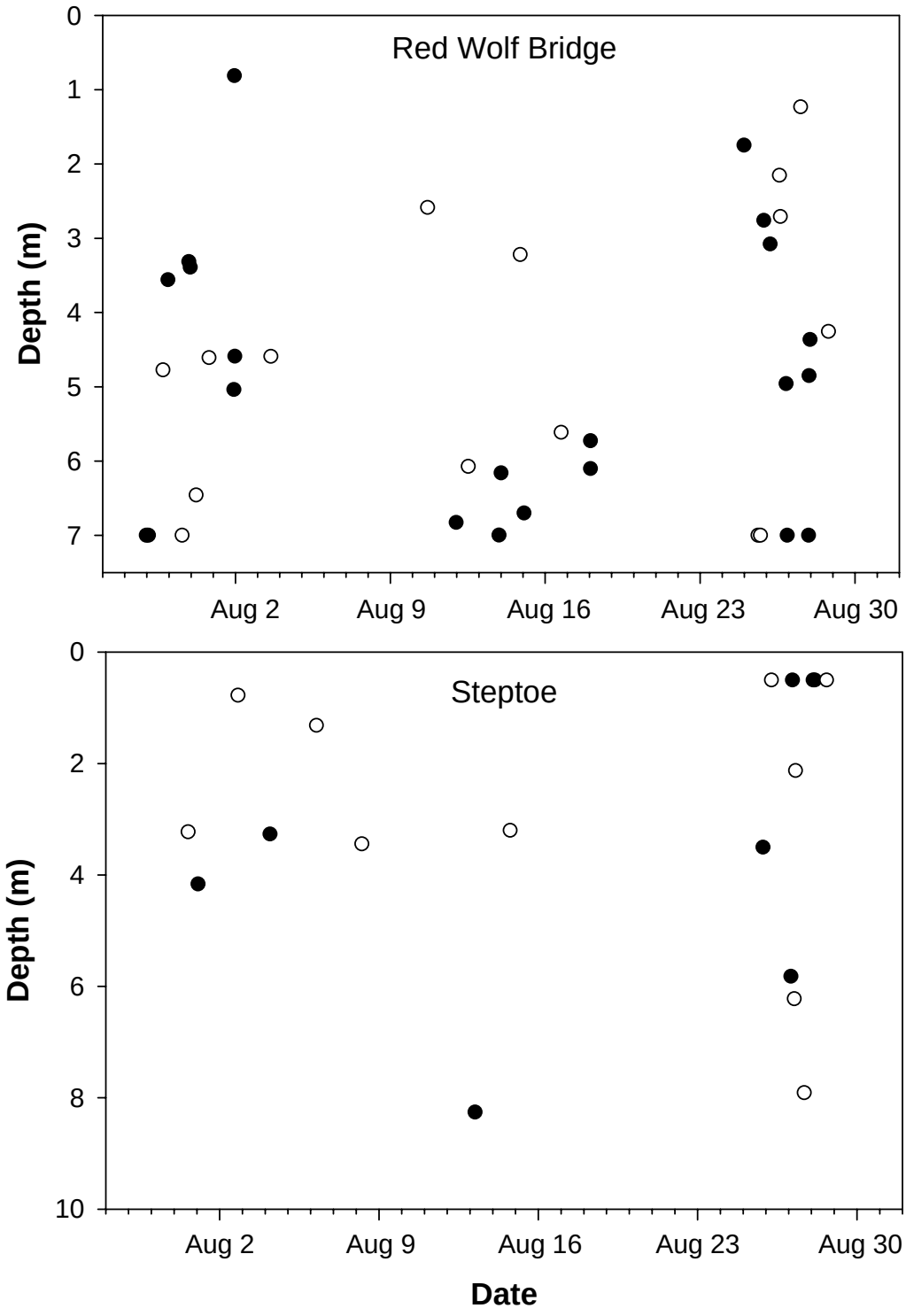


Figure 16.─Depths of juvenile fish passing Red Wolf Bridge (top panel) and Steptoe (bottom panel) on the Snake River during the day (open circles) and night (solid circles) in 2004.

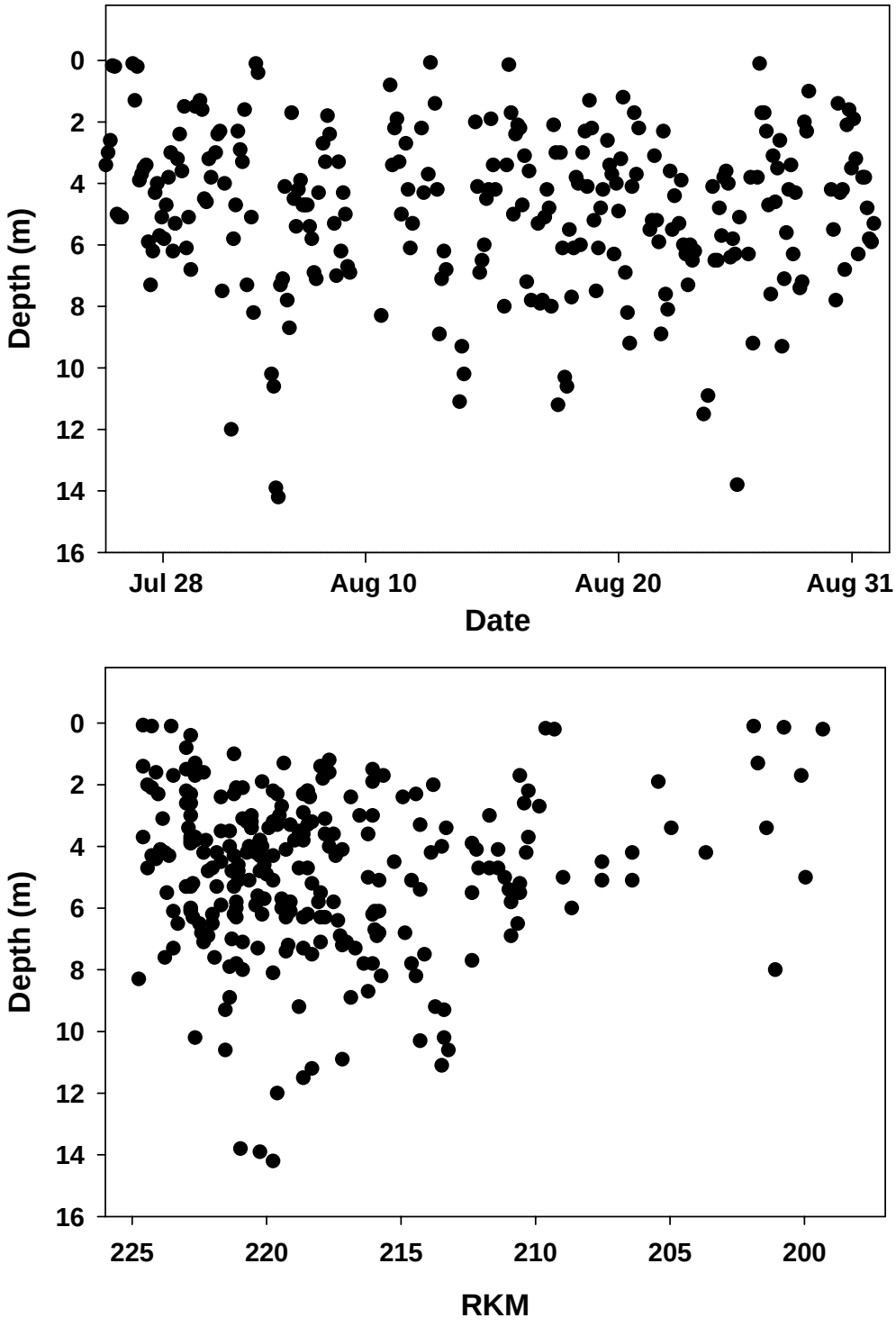


Figure 17.—Depths of subyearling fall Chinook salmon detected during mobile tracking in Lower Granite Reservoir, 2004. The top panel shows the depths used during the season and the bottom panel shows the depths used as fish moved downstream from their release at the confluence of the Snake and Clearwater rivers.

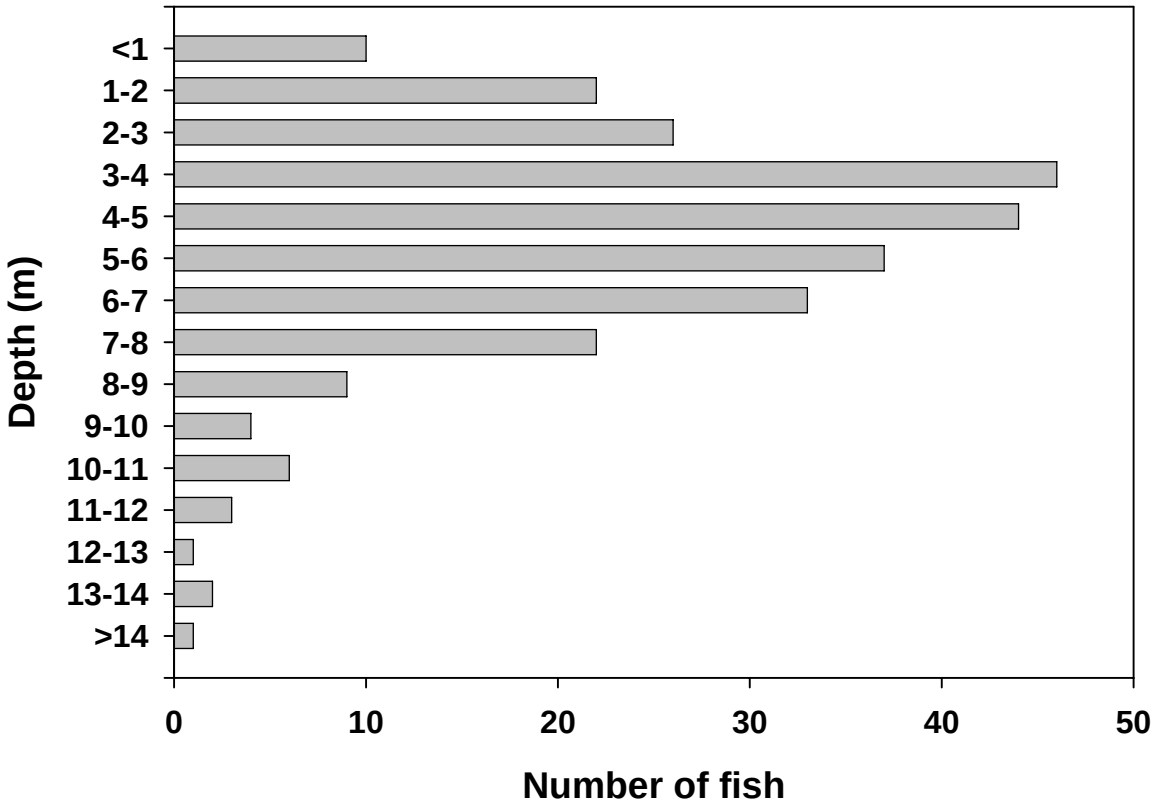


Figure 18.—Frequency of depths used by all subyearling fall Chinook salmon detected during mobile tracking in Lower Granite Reservoir, 2004.

Discussion

One of the benefits of using water released from Dworshak Reservoir for summer flow augmentation is the cooling effect it has on Lower Granite Reservoir. The thermal characteristics of Lower Granite Reservoir that result from summer flow augmentation have been well described by Cook et al. (2006). Without summer flow augmentation, Connor et al. (2001; Chapter One this report) showed that temperatures in Lower Granite Reservoir would exceed the 20°C temperature standard for the Lower Snake River and at times would approach the upper incipient lethal temperature for juvenile Chinook salmon. Our water temperature monitoring showed that cool water exists in Lower Granite Reservoir that could provide a thermal refuge for juvenile fall Chinook salmon. However, for summer flow augmentation to be effective, fish must be able to locate and use this cooler water.

We showed that juvenile fall Chinook salmon released at the confluence of the Snake and Clearwater rivers select a range of water temperatures indicating that fish can use cooler water if they choose. The Snake River above the confluence with the Clearwater River often exceeded 22°C during much of July and August and provides no thermal refuge. Fish reaching the confluence can select temperatures ranging from about 14-24°C. However as mixing of warm and cool water occurs in a downstream direction, this temperature range becomes narrower as deeper waters become warmer. Consequently, fish occupied a more restricted range of temperatures as they moved downstream, but most fish stayed in 17-18°C water.

The migration rates we observed for subyearlings in 2004 were similar to those we observed in 2002, 2003, and 2005 (Tiffan et al. 2003, 2005; USGS unpublished data) indicating fish behavior was not adversely affected by the radio tag or release location. Fish were more active at night than during the day and tended to occupy cooler temperatures during periods of inactivity. Beeman and Maule (2001) observed higher passage of juvenile Chinook salmon at McNary Dam at night indicating that fish are potentially more active at night. It is possible that temperature selection is less important to fish during periods of active migration although examination of individual thermal histories (Appendix 1) show that fish use a variety of temperatures during active movement.

The current temperature standard for the lower Snake River adopted by the state of Washington is 20°C. Most fish we observed selected temperatures less than 20°C as they moved downstream from the confluence. Little information currently exists on the long-term consequences of exposure to temperatures >20°C. Yankee (2005) observed the highest growth rate in subyearlings at 20°C in laboratory experiments but found reduced growth and mortality at 24°C. However, Marine and Cech (2004found reduced growth rates at temperatures from 21-24°C in laboratory experiments using fall-run Chinook salmon from the Sacramento River. We speculate that 20°C is near the peak of the temperature-growth curve and temperature >20°C may represent the descending limb of the curve where metabolic costs exceed energy intake in the presence of unlimited food. Temperatures above 20°C may also pose thermal challenges for subyearling fall Chinook salmon that may manifest themselves in increased predation and reduced disease resistance (Marine and Cech 2004). The cooler water provided by summer flow augmentation from Dworshak Reservoir may increase subyearling survival by reducing these negative effects. However, the range of cooler temperatures provided by flow augmentation is

only available to subyearlings in Lower Granite Reservoir. In reservoirs downstream of Lower Granite Dam, the range in temperatures is much reduced (Cook et al. 2006) and fish show no temperature selection in these relatively homogenous thermally environments (Tiffan et al. 2003).

Our conclusions are limited in part by the radio tags we used in this study. The size of the tag restricted us to only tagging relatively large fish late in the outmigration. In addition, the short life span of the tag did not enable us to track fish throughout the entire reservoir, which would have provided more complete thermal histories for tagged fish. It is possible that the variation in temperatures used in the upper reservoir may become reduced as both the range of available temperatures is restricted due to mixing and fish have more time to select a preferred temperature. An additional study limitation was our inability to determine the true availability of temperatures to fish, which would have enabled us to conduct selection analyses. Because fish are moving in three dimensions through time, temperature availability could only be determined using a time-varying three-dimensional temperature model which was beyond the scope of this study, but represents a future area of research.

The temperature selection displayed by subyearlings in this study provides a mechanism for the expression of a reservoir-type life history in Lower Granite Reservoir (Connor et al. 2005). The cooler water provided by summer flow augmentation may provide suitable conditions for late migrants to delay their seaward migration and spend the summer and winter in Lower Granite Reservoir. This may be particularly true for migrants originating in the Clearwater River, whose acclimation to cooler Clearwater River temperatures may cause an avoidance of warmer temperatures in Lower Granite Reservoir that may lead to migratory delay. However, this hypothesis remains unexplored and represents an important area of future research.

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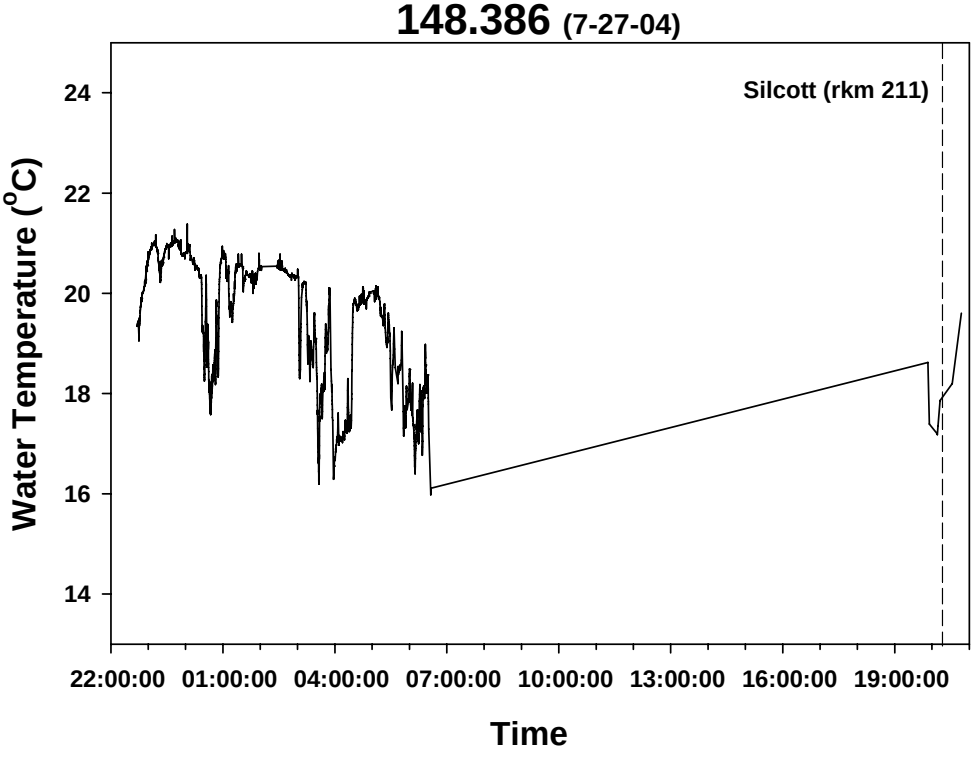
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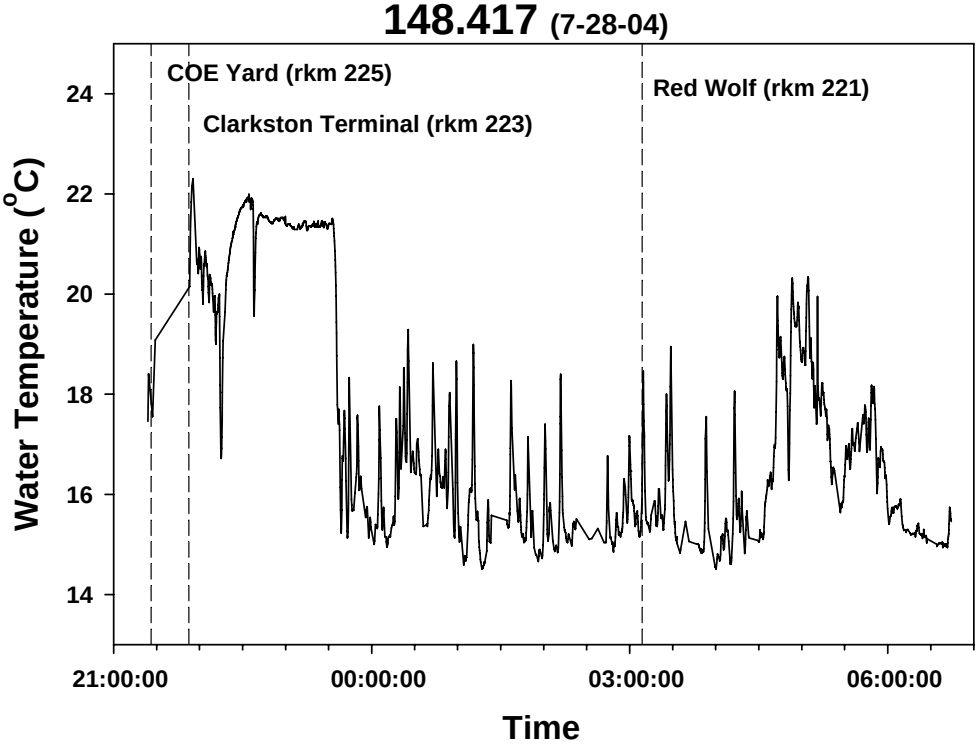
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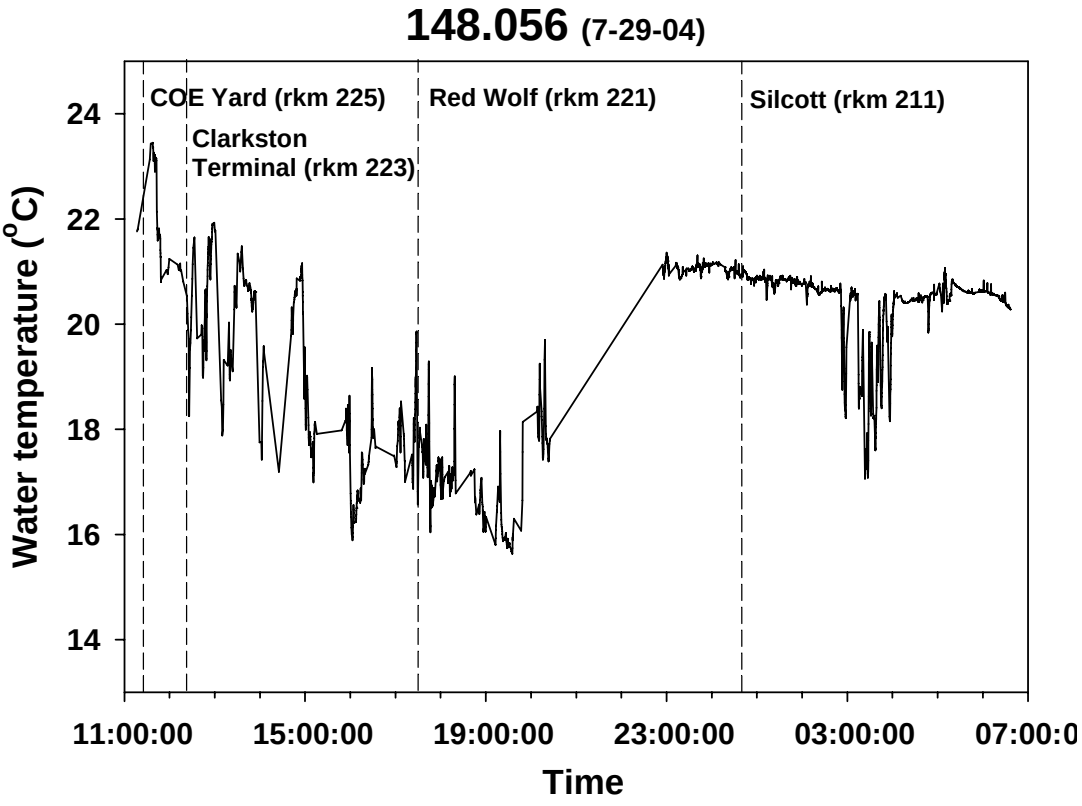
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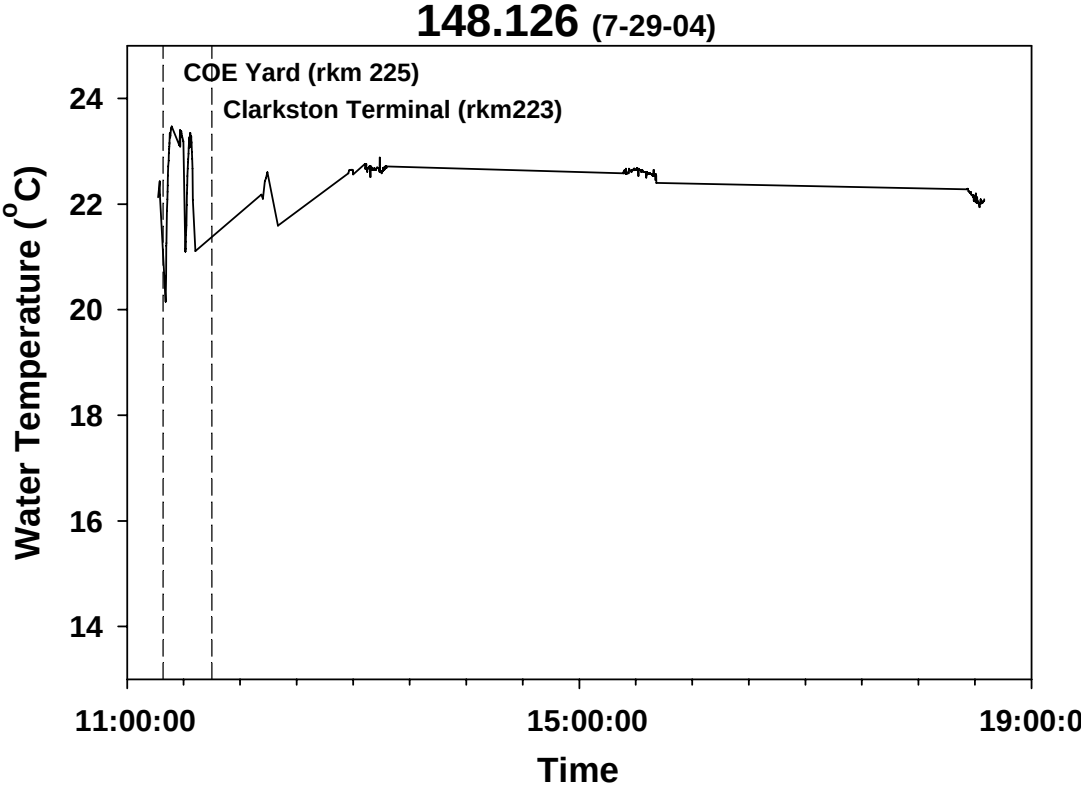
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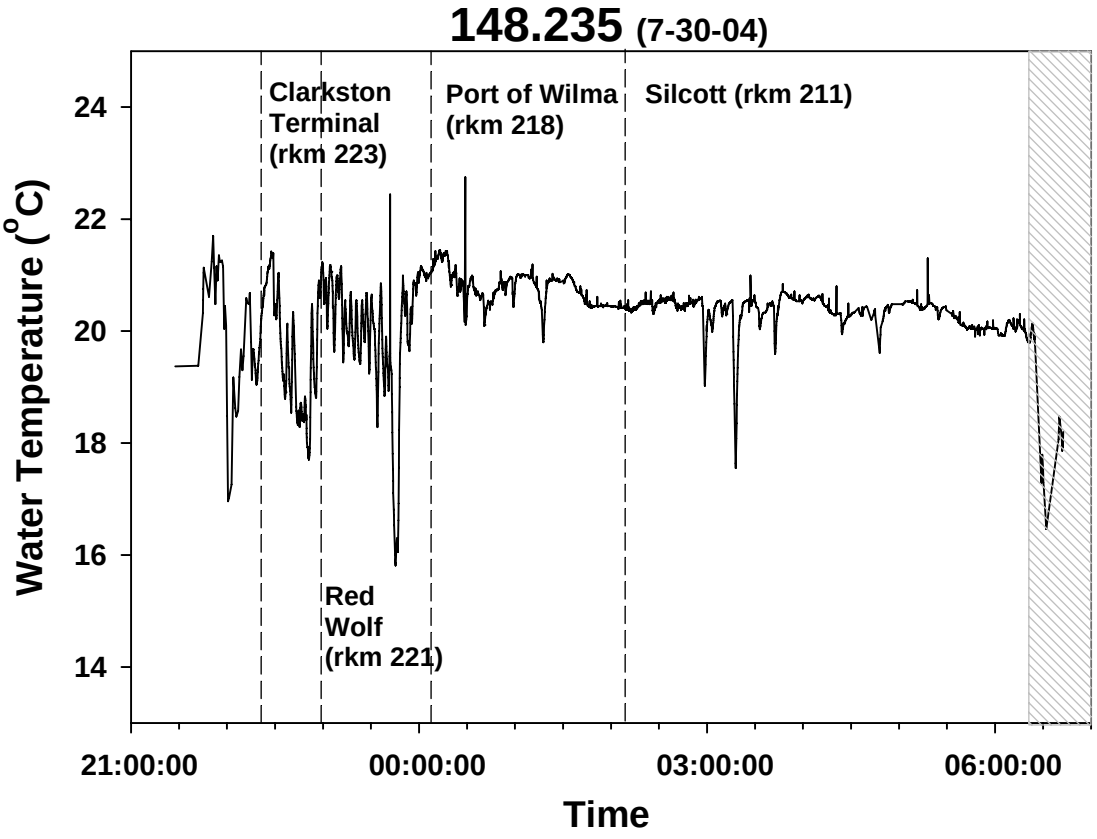
Appendix 1.-Plots of individual subyearling fall Chinook salmon thermal histories in the Snake River, 2004. The frequency of the temperature-sensing radio tag is shown at the top of each panel. Vertical dashed lines and locations are provided for reference. Shaded areas represent times of inactivity.

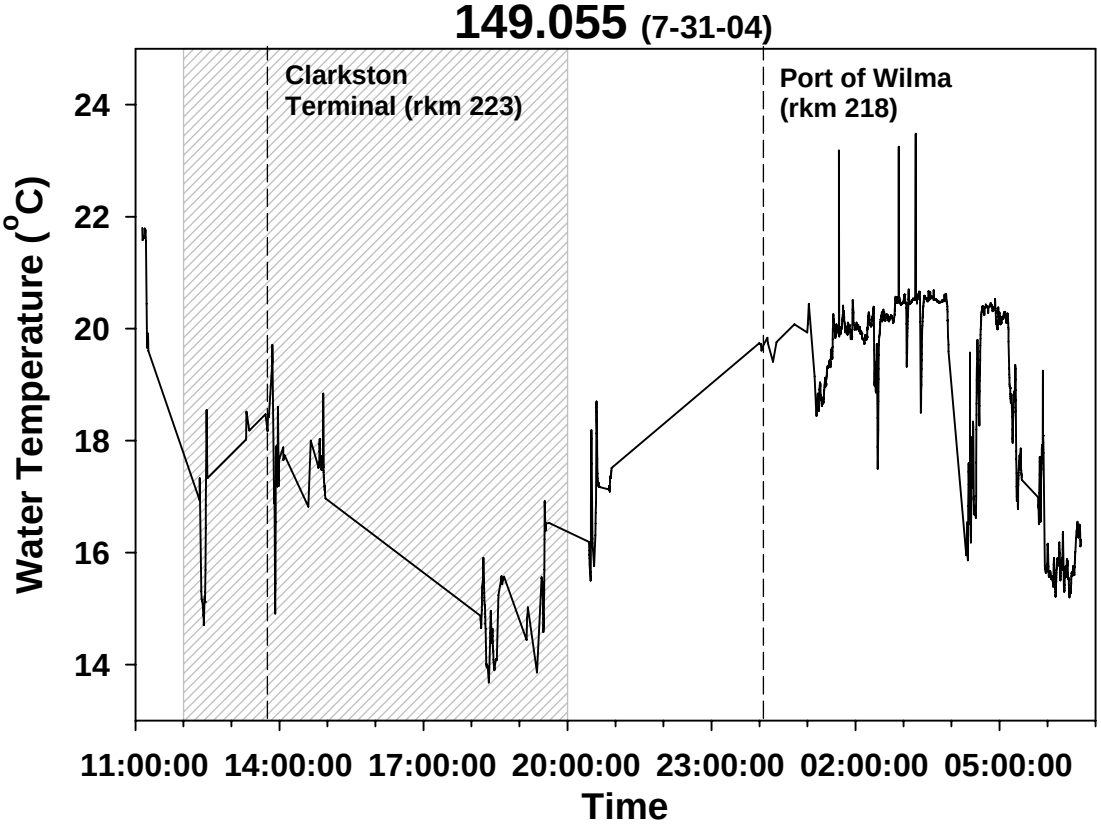


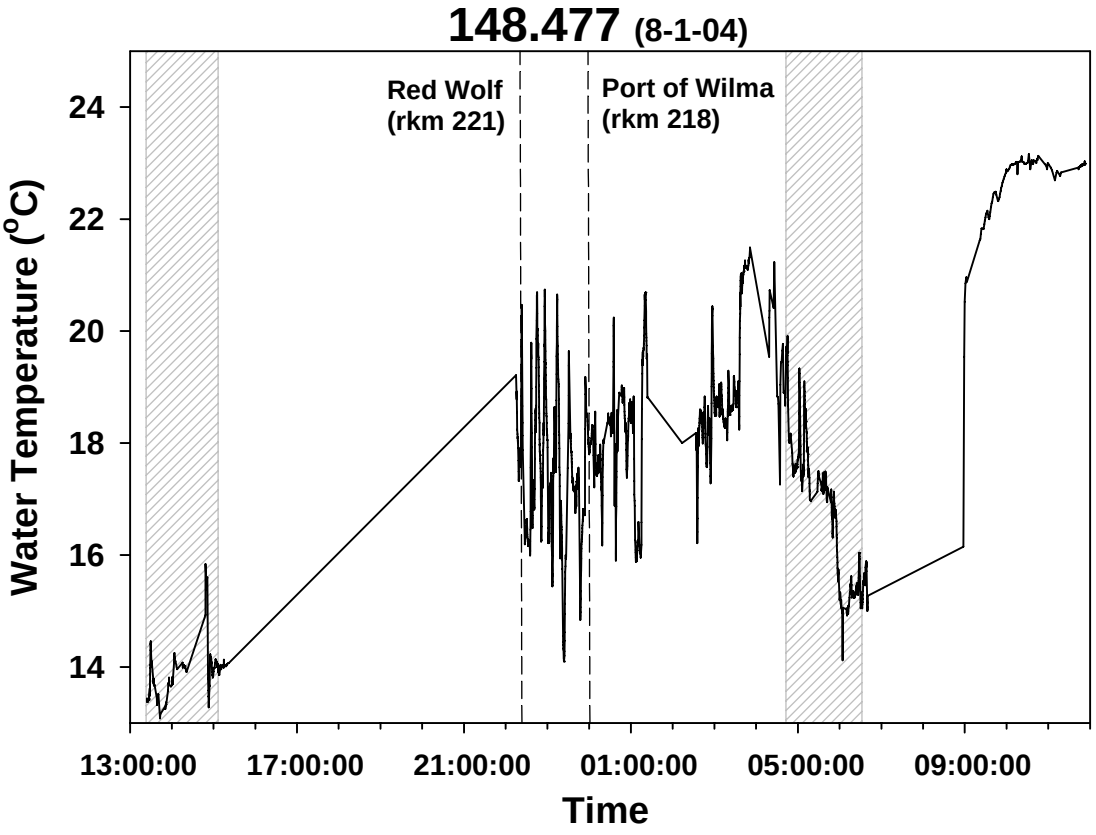


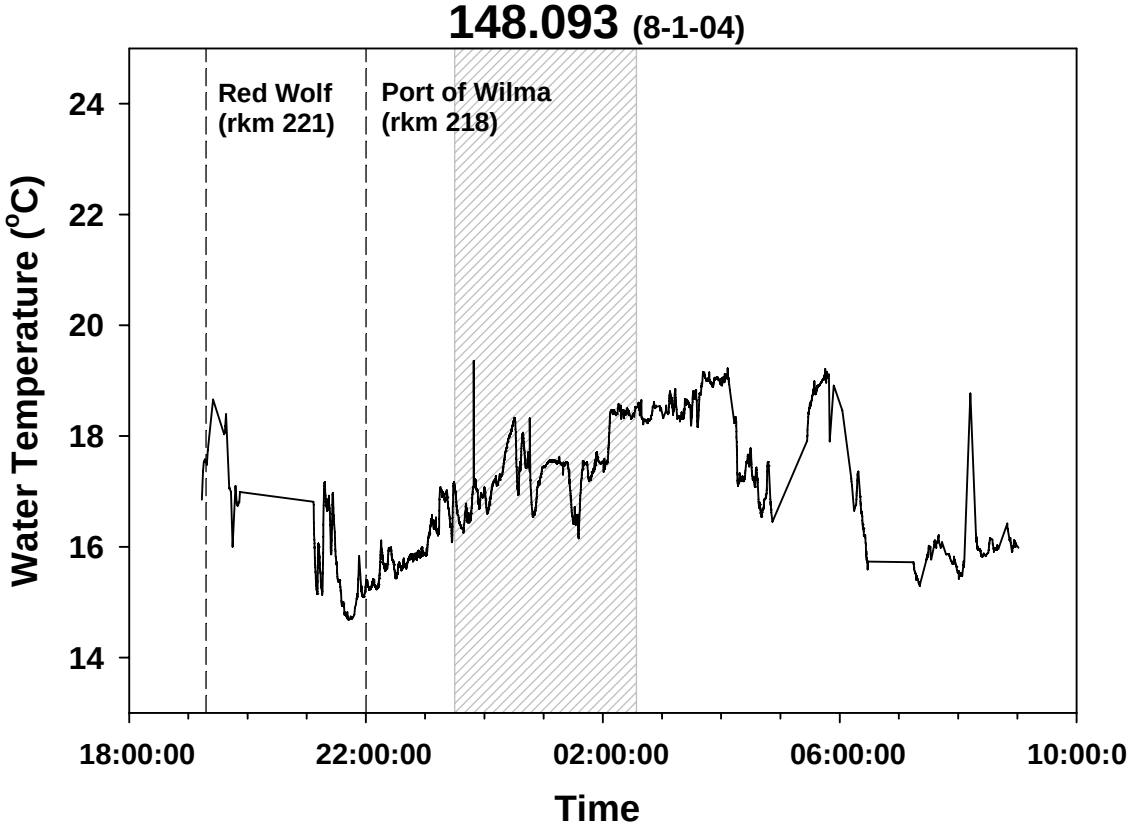

148.056 (7-29-04)
COE Yard (rkm 225)
Clarkston Terminal (rkm 223)
Red Wolf (rkm 221)
Silcott (rkm 211)
Water temperature (°C)
24
22
20
18
16
14
11:00:00
15:00:00
19:00:00
23:00:00
03:00:00
07:00:0
Time

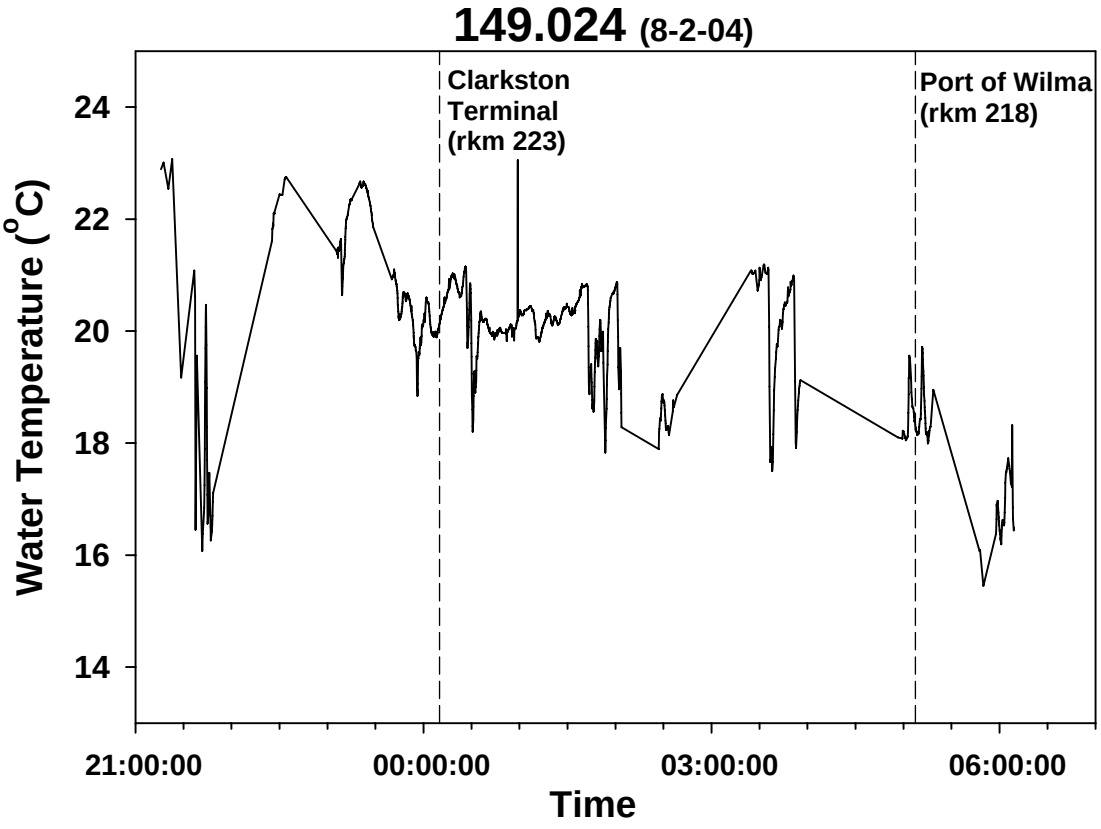

148.126 (7-29-04)
COE Yard (rkm 225)
Clarkston Terminal (rkm223)
Water Temperature (°C)
24
22
20
18
16
14
11:00:00
15:00:00
19:00:0
Time

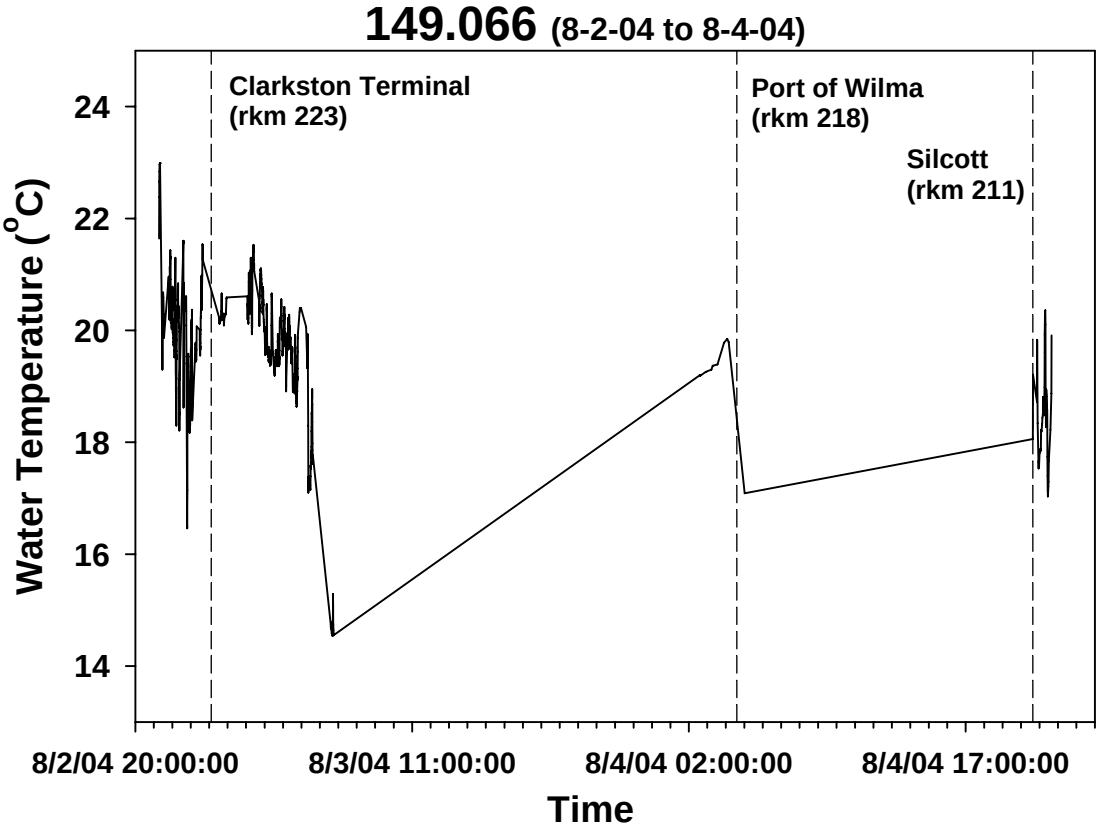

148.235 (7-30-04)
Clarkston Terminal (rkm 223)
Port of Wilma (rkm 218)
Silcott (rkm 211)
Red Wolf (rkm 221)
Water Temperature (°C)
24
22
20
18
16
14
21:00:00
00:00:00
03:00:00
06:00:00
Time

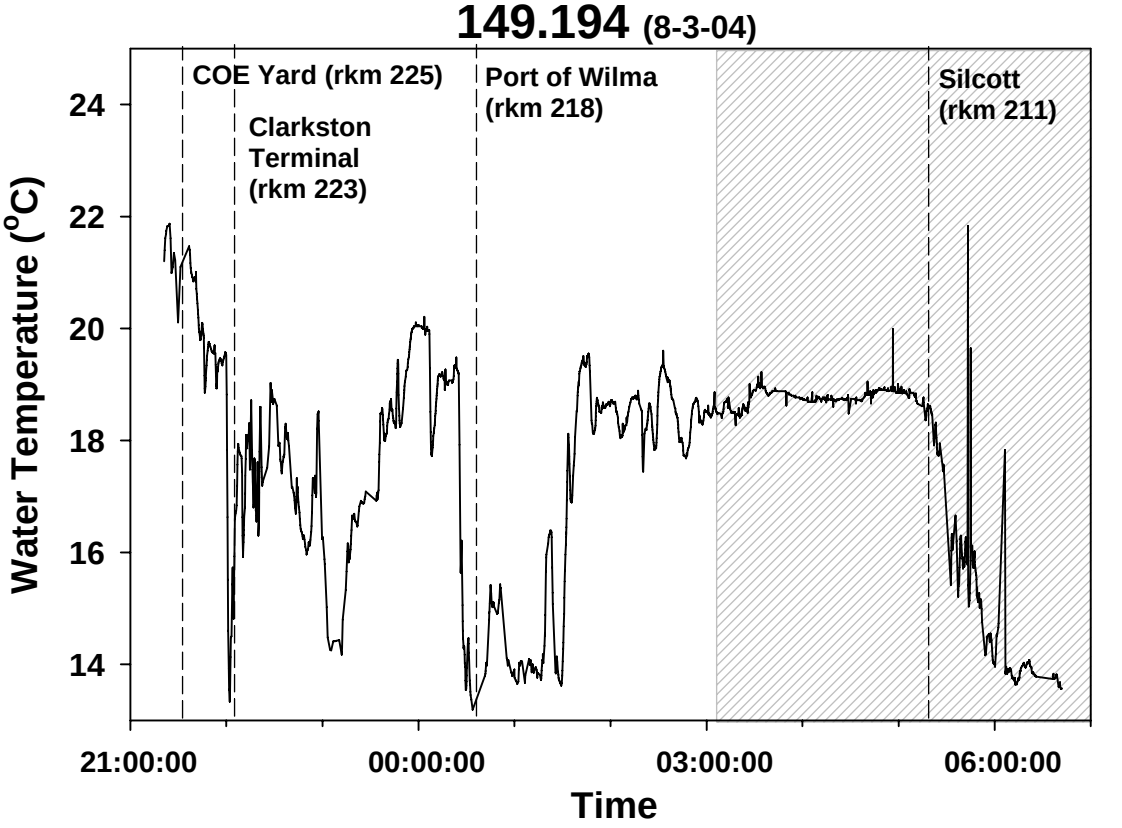

149.055 (7-31-04)
Clarkston Terminal (rkm 223)
Port of Wilma (rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
11:00:00
14:00:00
17:00:00
20:00:00
23:00:00
02:00:00
05:00:00
Time

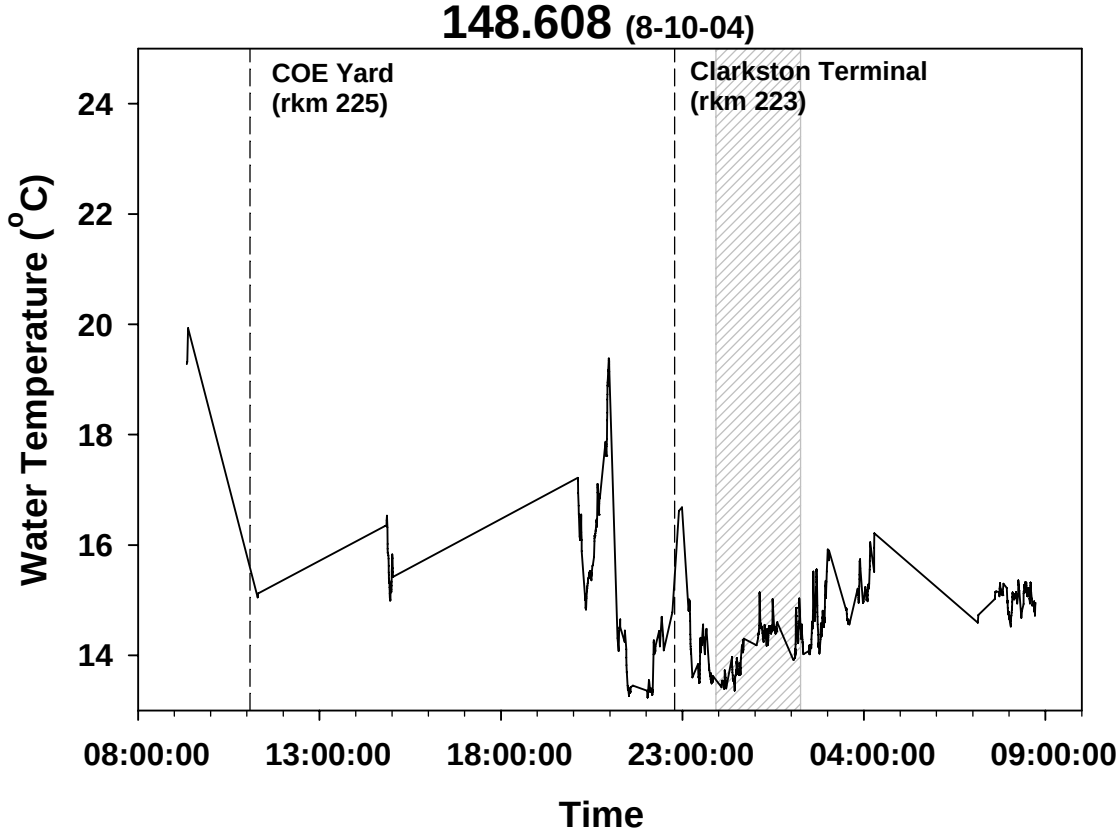

148.477 (8-1-04)
Red Wolf
(rkm 221)
Port of Wilma
(rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
13:00:00
17:00:00
21:00:00
01:00:00
05:00:00
09:00:00
Time

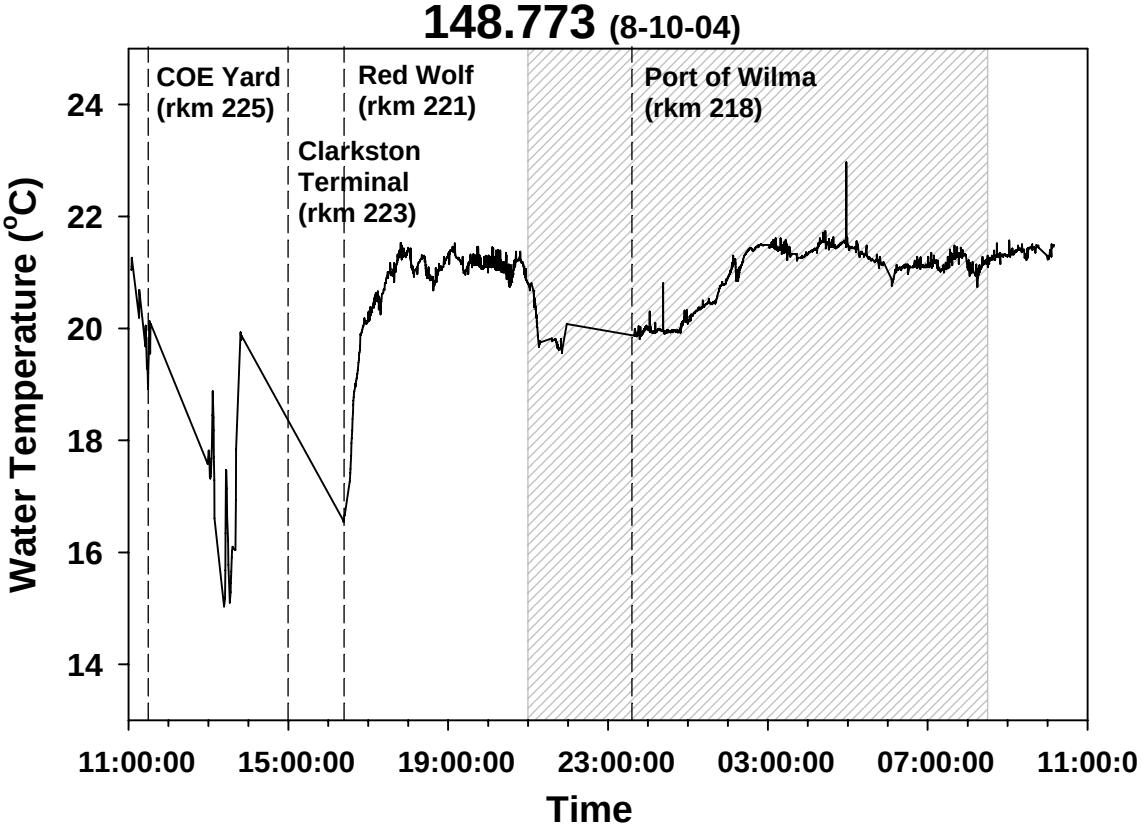

148.093 (8-1-04)
Red Wolf
(rkm 221)
Port of Wilma
(rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
18:00:00
22:00:00
02:00:00
06:00:00
10:00:0
Time

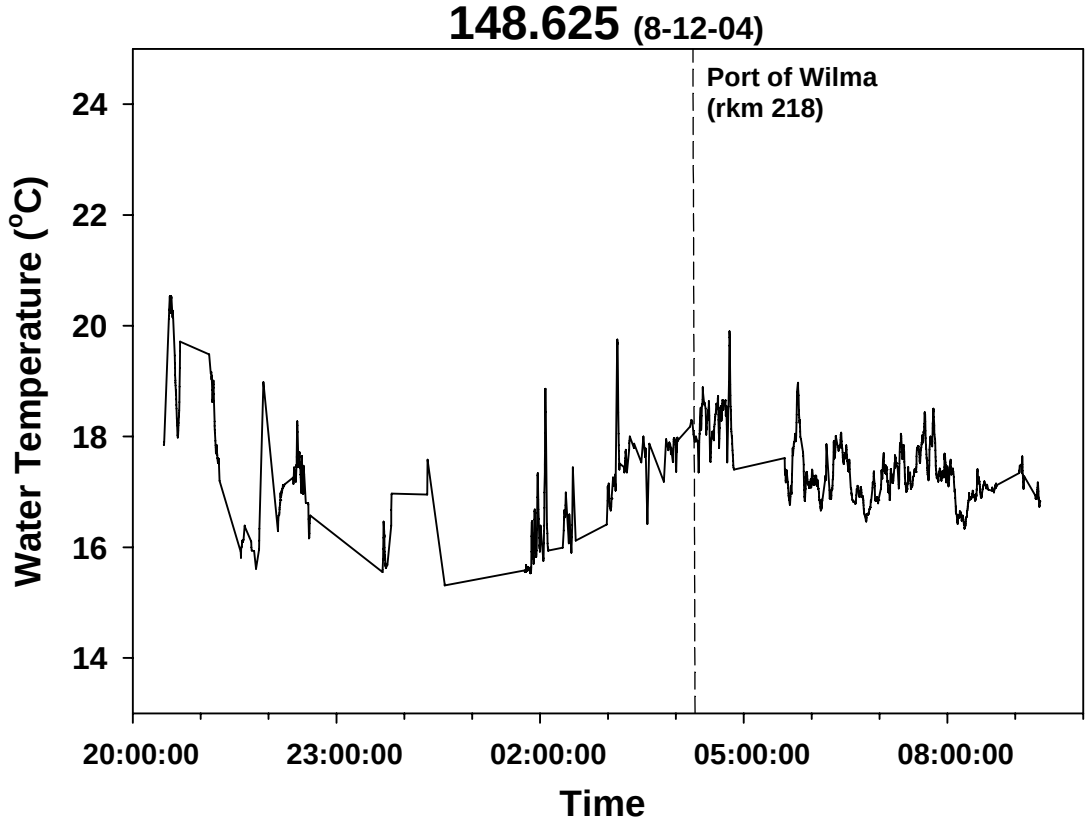

149.024 (8-2-04)
Clarkston Terminal (rkm 223)
Port of Wilma (rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
21:00:00
00:00:00
03:00:00
06:00:00
Time

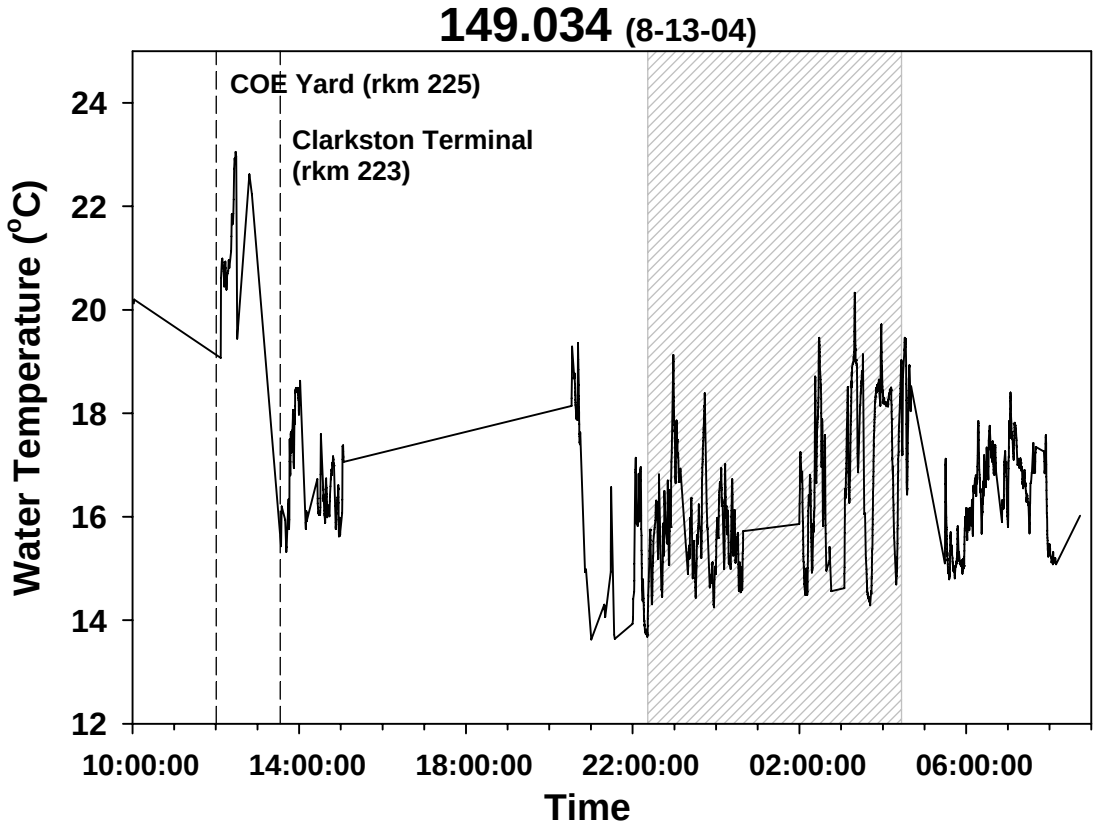

149.066 (8-2-04 to 8-4-04)
Clarkston Terminal (rkm 223)
Port of Wilma (rkm 218)
Silcott (rkm 211)
Water Temperature (°C)
24
22
20
18
16
14
8/2/04 20:00:00
8/3/04 11:00:00
8/4/04 02:00:00
8/4/04 17:00:00
Time

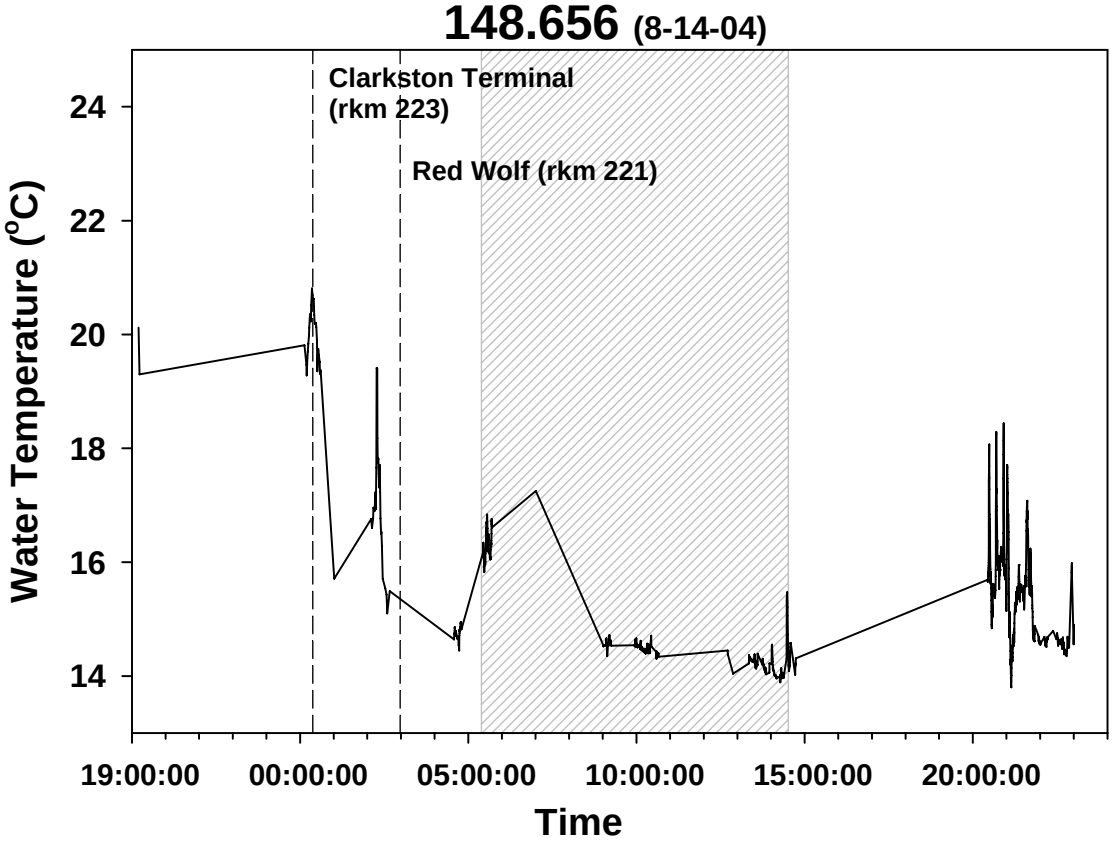

149.194 (8-3-04)
COE Yard (rkm 225)
Clarkston Terminal (rkm 223)
Port of Wilma (rkm 218)
Silcott (rkm 211)
Water Temperature (°C)
24
22
20
18
16
14
21:00:00
00:00:00
03:00:00
06:00:00
Time

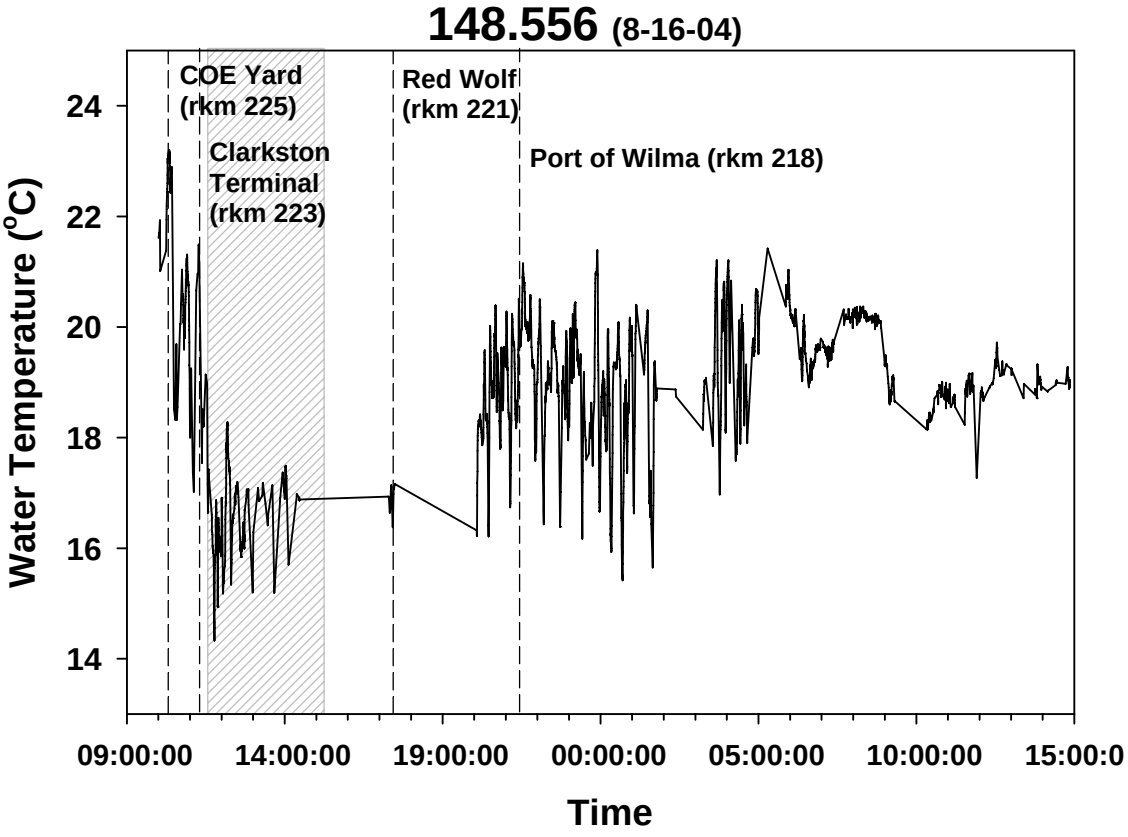

148.608 (8-10-04)
COE Yard (rkm 225)
Clarkston Terminal (rkm 223)
Water Temperature (°C)
24
22
20
18
16
14
08:00:00
13:00:00
18:00:00
23:00:00
04:00:00
09:00:00
Time

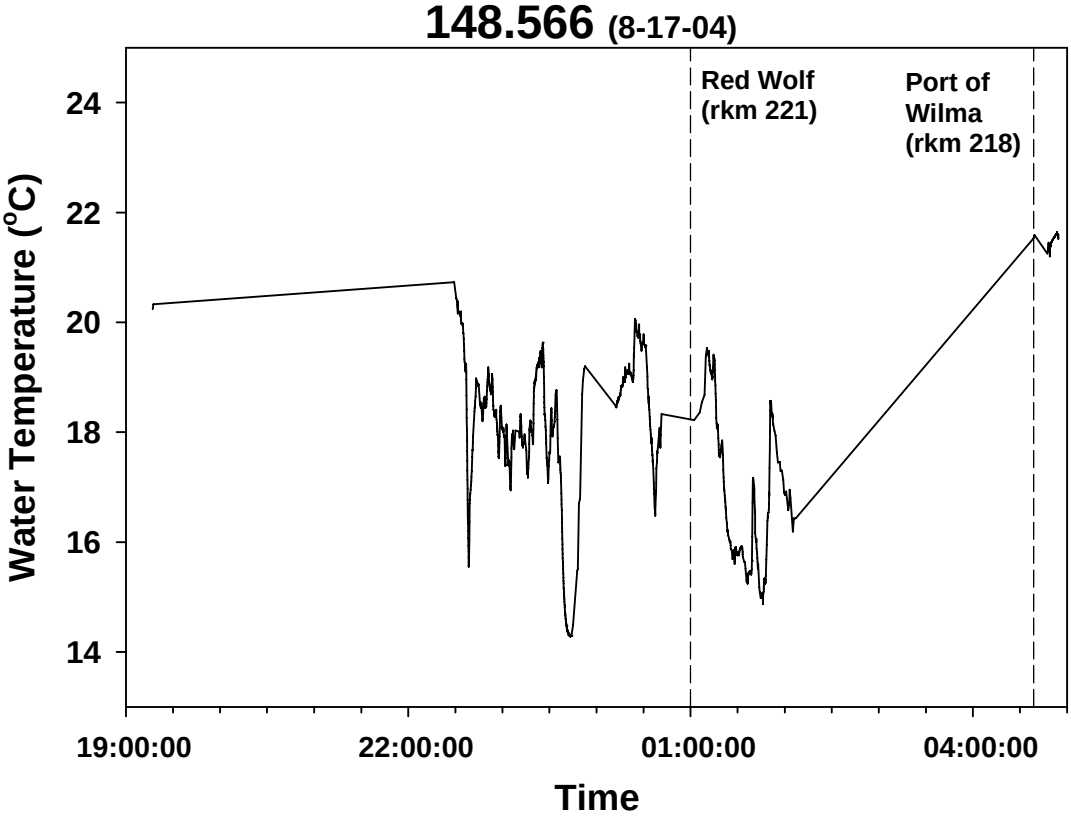

148.773 (8-10-04)
COE Yard
(rkm 225)
Red Wolf
(rkm 221)
Port of Wilma
(rkm 218)
Clarkston
Terminal
(rkm 223)
Water Temperature (°C)
24
22
20
18
16
14
11:00:00
15:00:00
19:00:00
23:00:00
03:00:00
07:00:00
11:00:0
Time

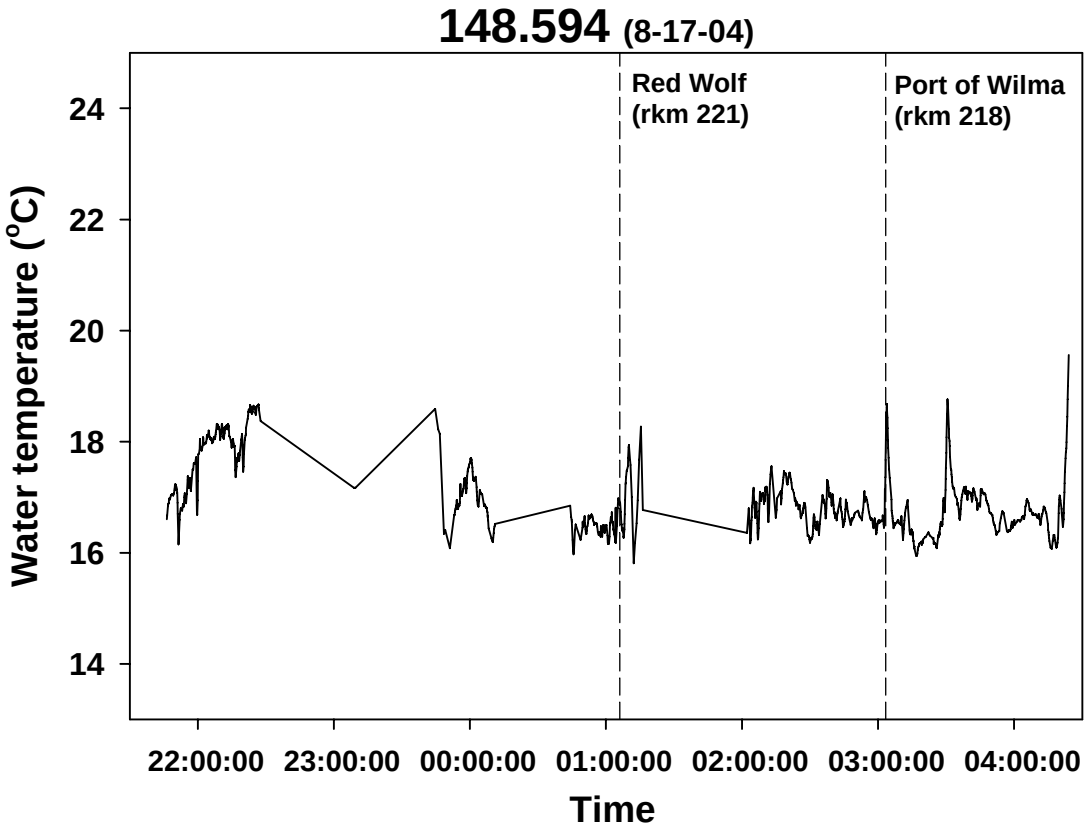

148.625 (8-12-04)
Port of Wilma
(rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
20:00:00
23:00:00
02:00:00
05:00:00
08:00:00
Time

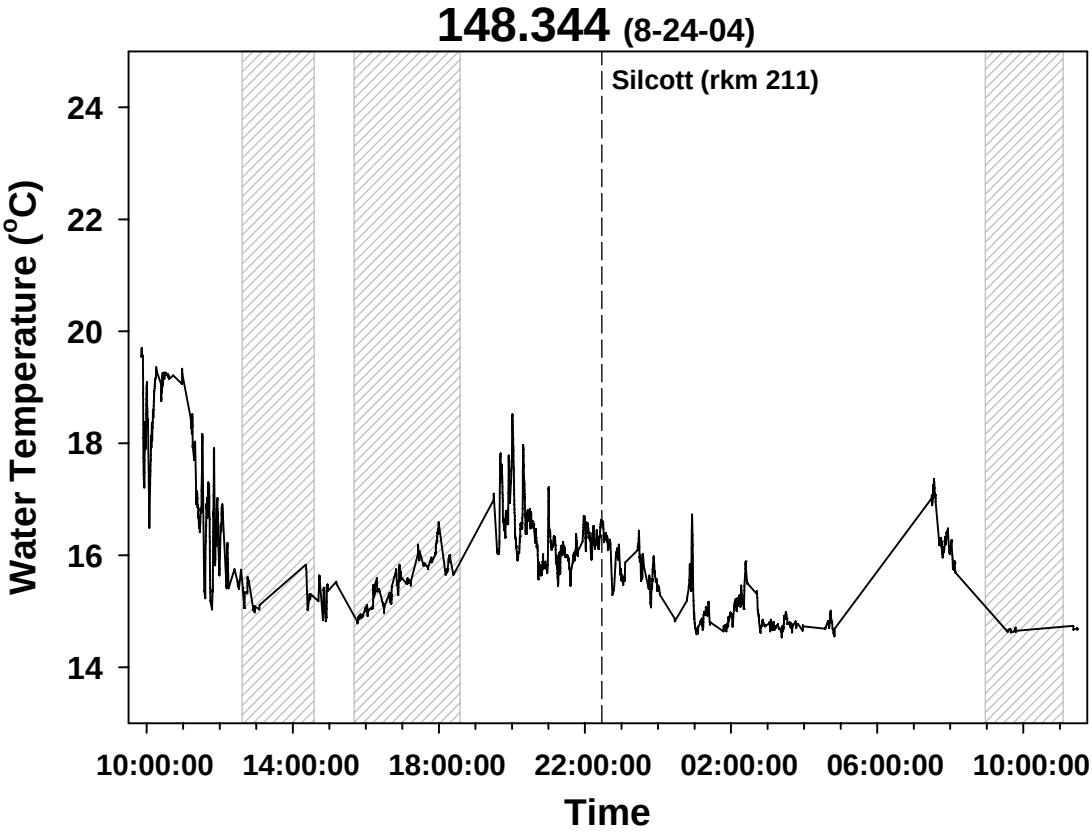

149.034 (8-13-04)
COE Yard (rkm 225)
Clarkston Terminal
(rkm 223)
Water Temperature (°C)
24
22
20
18
16
14
12
10:00:00
14:00:00
18:00:00
22:00:00
02:00:00
06:00:00
Time

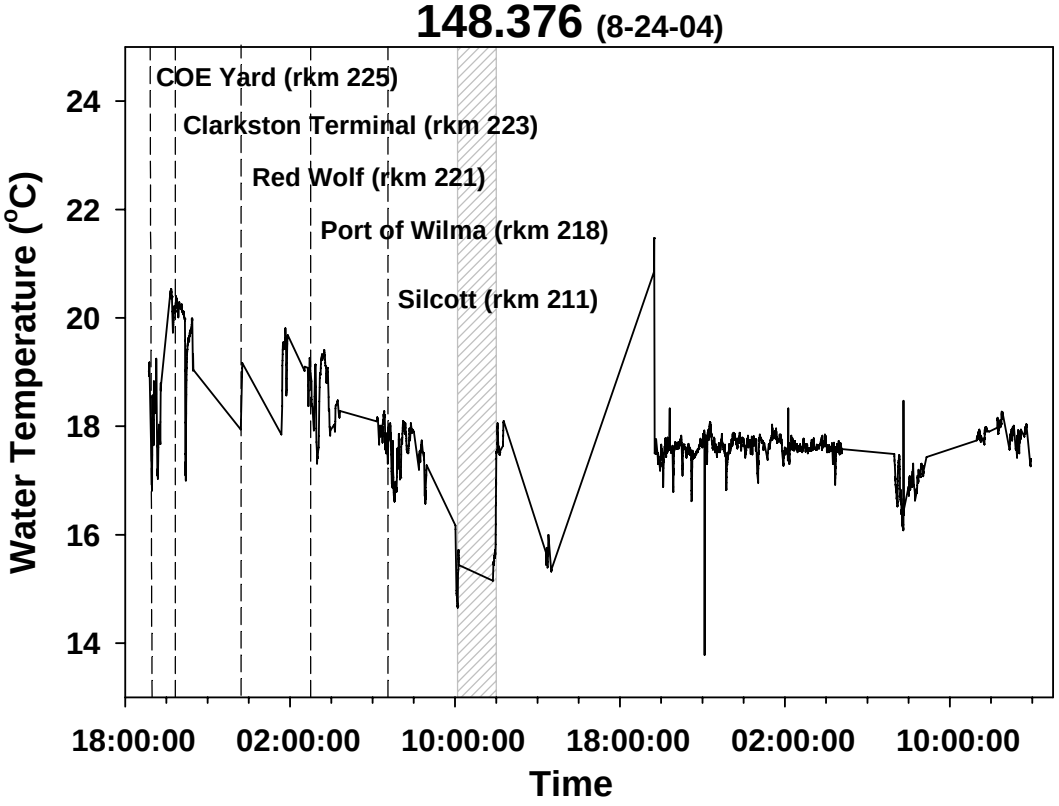

148.656 (8-14-04)
Clarkston Terminal
(rkm 223)
Red Wolf (rkm 221)
Water Temperature (°C)
24
22
20
18
16
14
19:00:00
00:00:00
05:00:00
10:00:00
15:00:00
20:00:00
Time

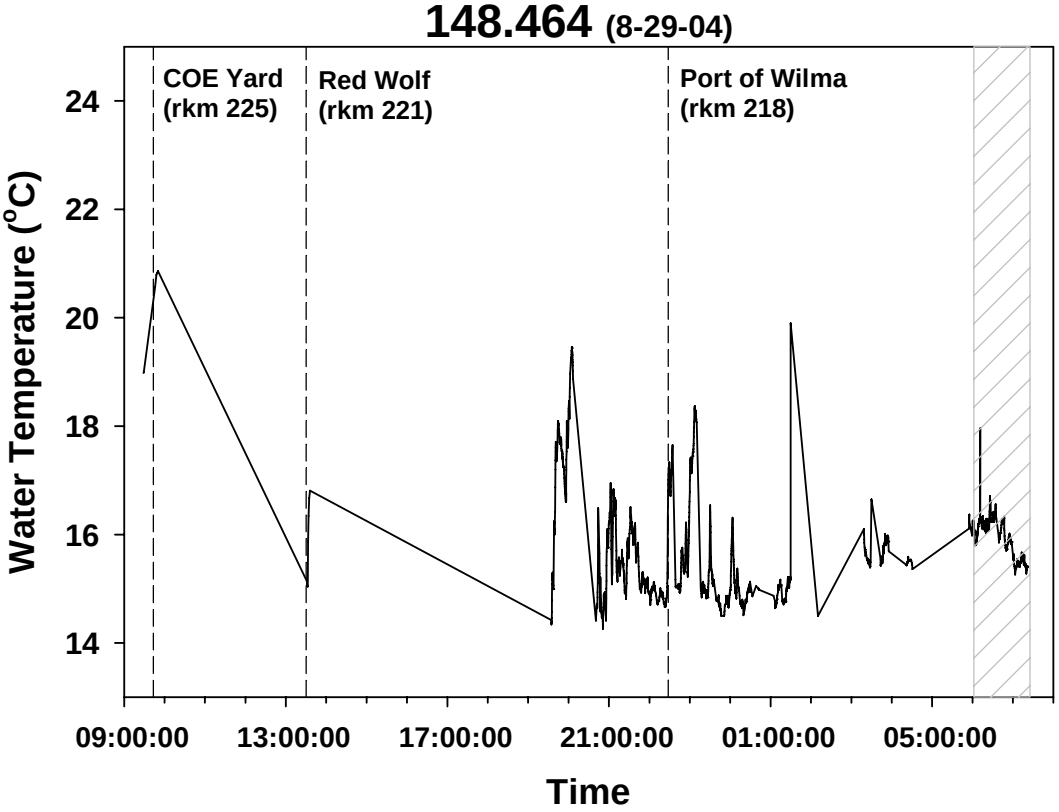

148.556 (8-16-04)
COE Yard (rkm 225)
Clarkston Terminal (rkm 223)
Red Wolf (rkm 221)
Port of Wilma (rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
09:00:00
14:00:00
19:00:00
00:00:00
05:00:00
10:00:00
15:00:0
Time

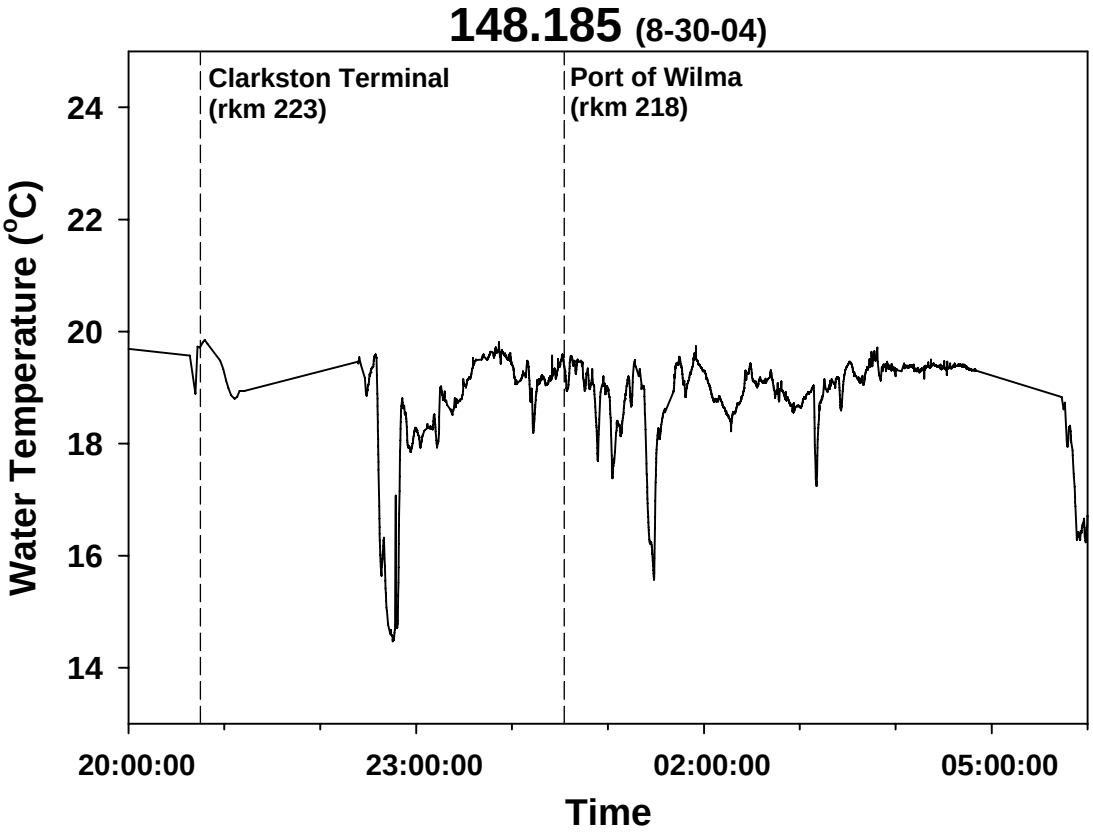

148.566 (8-17-04)
Red Wolf (rkm 221)
Port of Wilma (rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
19:00:00
22:00:00
01:00:00
04:00:00
Time


148.594 (8-17-04)
Red Wolf
(rkm 221)
Port of Wilma
(rkm 218)
Water temperature (°C)
24
22
20
18
16
14
22:00:00
23:00:00
00:00:00
01:00:00
02:00:00
03:00:00
04:00:00
Time


148.344 (8-24-04)
Silcott (rkm 211)
Water Temperature (°C)
24
22
20
18
16
14
10:00:00
14:00:00
18:00:00
22:00:00
02:00:00
06:00:00
10:00:00
Time


148.376 (8-24-04)
COE Yard (rkm 225)
Clarkston Terminal (rkm 223)
Red Wolf (rkm 221)
Port of Wilma (rkm 218)
Silcott (rkm 211)
Water Temperature (°C)
24
22
20
18
16
14
18:00:00
02:00:00
10:00:00
18:00:00
02:00:00
10:00:00
Time


148.464 (8-29-04)
COE Yard (rkm 225)
Red Wolf (rkm 221)
Port of Wilma (rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
09:00:00
13:00:00
17:00:00
21:00:00
01:00:00
05:00:00
Time


148.185 (8-30-04)
Clarkston Terminal
(rkm 223)
Port of Wilma
(rkm 218)
Water Temperature (°C)
24
22
20
18
16
14
20:00:00
23:00:00
02:00:00
05:00:00
Time

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